TCS CORPORATE WEBSITE

Cards Block — Developer Guide

Role / audience	Development — Developers, technical leads, block owners, AI coding agents
Version	1.0
Status	Draft for review
Owner	Technical Lead / Block Owner (proposed)
Last validated	2026-11-08
Documents	1
Classification	INTERNAL — classification to be confirmed

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Contents

Cards Block — Developer Guide
- Summary
- Overview
- Source files
- Components, models and filters
- Fields
- Dependencies
- For Developers
- Extending the Block
- Responsive Behavior
- Accessibility
- Data Layer and Analytics
- Performance Considerations
- Security and Privacy
- For Architects
- Troubleshooting
- Known limitations of the block
- Change history of the block
- Reference pages
- Coverage of the required topics
- Screenshots
- Known limitations
- Open items
- Change history

Cards Block — Developer Guide

Attribute	Value
Document ID	DEV-B-cards
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How the Cards block is built: files, model, structure, behaviour, variations, tests and limitations
Scope	blocks/cards/ and its components cards, card
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	8 November 2026
Last validated	2026-11-08 (model, files, imports from the code; reference pages from the live site)
Source files	blocks/cards/README.md, blocks/cards/_cards.json, blocks/cards/cards-data.js, blocks/cards/cards.css, blocks/cards/cards.js
Related documents	Variations, Test coverage, Authoring guide
Related AEM paths	/content/tcs-corp/design-block-master/en/block-library/cards, /content/tcs-corp/design-block-master/en/block-library/cards/card-colours, /content/tcs-corp/design-block-master/en/block-library/cards/card-styles

1. Summary

Attribute	Value
Block	Cards
Folder	`blocks/cards`
Type	Container
Category	listing
Status	Active
Purpose	Manually authored cards with the List's presentation (grid or horizontal scroll)
Variations (summary)	Card style (image, panel, report, statistic, mixed), Card colour, layout per breakpoint
Children	card (Card)
Code size	282 lines of JS and CSS

2. Overview

- **Business purpose:** presents curated, hand-written content as consistent, brand-compliant cards without creating a page for every card: statistics ("4bn revenue"), capabilities, services described in campaign wording, partner logos with a text, report downloads, or promotions pointing to pages or

external sites. Because Cards uses the List's design and settings, a page can mix curated cards and page lists without visual differences.

- **User experience:** an optional header (title, description and a call to action on the right), then the cards in the authored order: image cards, coloured panels, report cards with a Download action, statistic cards with a large number, an optional full-width feature card above the others, or full-width rows of links with an arrow. Each breakpoint is a grid or a horizontal scroller with optional previous / next buttons and a scroll indicator.
- **Templates / page types:** any page; no page template adds or removes it. Typical placements: home and landing pages (statistics, capabilities), article and blog pages (related content curated by hand), press-release pages (downloads).
- **Data source:** authored fields only. Tag labels come from the locale's taxonomy (`/{locale}/taxonomy.json`) and UI text from its placeholders. Cards reads no page metadata, query index, redirects sheet or Authors sheet.
- **Related blocks:** List (the same presentation for pages: child pages, fixed list, tagged pages); Filter List (page discovery with filter pills and sorting); Teaser and Hero (single promotions); Table List (text-led lists without images); Columns Container (free side-by-side layouts).

	Cards	List
Content	Typed in each card	The pages' metadata (card title, description, image, ...), with overrides per item (Fixed list)
Destinations	Any page or URL, or none (a card without a link)	Pages of the site or URLs
Updates	Edited in the block	Follow the pages (a new child page or tagged page shows up)
Rich text	Description is rich text (paragraphs, bold, lists, inline links)	Plain text
Statistics, themes	Typed numbers and labels	The same, per item

- **When to use:** curated content that isn't a page of its own or whose wording is written for this placement (statistics, capabilities, services with their own wording, logos with text, external promotions).
- **When not to use:** cards that represent pages that already have card metadata (use the List, so the cards stay in sync with the pages); filterable or sortable listings (use Filter List); tabular data (Table or Table List); a single promotion (Teaser).
- **Realistic use cases:** an "At a glance" statistic row on an about page; a services rail with images, descriptions and "Learn more" calls to action; dark themed panels linking to topic hubs; a featured spotlight card followed by three smaller cards; report download teasers; a news-style list of links that isn't driven by the query index; Block Collection cards authored before the redesign, which keep rendering in the new design.

3. Source files

File	Link
`README.md`	GitHub
`_cards.json`	GitHub
`cards-data.js`	GitHub
`cards.css`	GitHub
`cards.js`	GitHub

4. Components, models and filters

Component	Title	Resource type	Template name	Model	Children allowed
cards	Cards	core/franklin/components/block/v1/block	Cards	cards	card
card	Card	core/franklin/components/block/v1/block/item	Card	card	—

5. Fields

The Additional tab fields (anchor, hide, padding, margin, animation) are common to the blocks and described in the content modeling guide.

5.1 Cards (cards)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Title	title	text	No	—	—	—
Content	Heading type	titleType	select	No	h2	5 options	—
Content	Description	description	richtext	No	—	—	—
Content	CTA link	link	aem-content	No	—	—	—
Content	CTA text	linkText	text	No	—	—	—
Content	CTA style	linkType	select	No	""	3 options	—
Content	Accessible list label	listLabel	text	No	—	—	—
Display	Display as teaser	displayAsTeaser	boolean	No	true	—	—
Display	Card style	cardStyle	select	No	image	5 options	Conditional
Display	Card colour	cardTheme	select	No	""	9 options	Conditional
Display	Image ratio	imageRatio	select	No	16-9	4 options	Conditional
Display	Panel height	panelHeight	select	No	portrait	3 options	Conditional
Display	First card as feature	featureFirst	boolean	No	false	—	Conditional
Display	Link items	linkItems	boolean	No	true	—	—
Display	Open links in a new tab	openInNewTab	boolean	No	false	—	—
Display	Show image	showImage	boolean	No	true	—	—
Display	Show eyebrow	showEyebrow	boolean	No	true	—	—
Display	Show tags	showTags	boolean	No	true	—	—
Display	Show description	showDescription	boolean	No	true	—	—

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Display	Show date	showDate	boolean	No	true	—	—
Display	Date format	dateFormat	select	No	dd-mmm-yy	6 options	Conditional
Display	Show call to action	showCta	boolean	No	true	—	—
Display	Image loading	imageLoading	select	No	lazy	2 options	—
Display	Desktop display	desktopMode	select	No	grid	2 options	—
Display	Desktop columns	desktopColumns	number	No	4	—	Conditional
Display	Desktop gap (px)	desktopGap	number	No	24	—	—
Display	Desktop items visible	desktopVisible	number	No	4.2	—	Conditional
Display	Desktop card width (px)	desktopCardWidth	number	No	0	—	Conditional
Display	Previous / next buttons on desktop	desktopControls	boolean	No	false	—	Conditional
Display	Scroll indicator on desktop	desktopIndicator	boolean	No	false	—	Conditional
Display	Tablet display	tabletMode	select	No	grid	2 options	—
Display	Tablet columns	tabletColumns	number	No	4	—	Conditional
Display	Tablet gap (px)	tabletGap	number	No	24	—	—
Display	Tablet items visible	tabletVisible	number	No	3.3	—	Conditional
Display	Tablet card width (px)	tabletCardWidth	number	No	0	—	Conditional
Display	Previous / next buttons on tablet	tabletControls	boolean	No	false	—	Conditional
Display	Scroll indicator on tablet	tabletIndicator	boolean	No	false	—	Conditional
Display	Mobile display	mobileMode	select	No	grid	2 options	—
Display	Mobile columns	mobileColumns	number	No	1	—	Conditional
Display	Mobile gap (px)	mobileGap	number	No	16	—	—
Display	Mobile items visible	mobileVisible	number	No	1.2	—	Conditional
Display	Mobile card width (px)	mobileCardWidth	number	No	0	—	Conditional
Display	Previous / next buttons on mobile	mobileControls	boolean	No	false	—	Conditional

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Display	Scroll indicator on mobile	`mobileIndicator`	boolean	No	`true`	—	Conditional
Display	Equal card height	`equalHeight`	boolean	No	`false`	—	—
Display	Stretch cards to row height	`stretchCards`	boolean	No	`true`	—	—
Display	Scroll snap	`scrollSnap`	boolean	No	`true`	—	Conditional
Display	Snap alignment	`snapAlign`	select	No	`start`	2 options	Conditional
Display	Hide native scrollbar	`hideScrollbar`	boolean	No	`true`	—	Conditional
Display	Allow mouse drag	`mouseDrag`	boolean	No	`true`	—	Conditional
Display	Keyboard scrolling	`keyboardScrolling`	boolean	No	`true`	—	Conditional
Display	Scroll step	`scrollStep`	select	No	`one-card`	2 options	Conditional
Display	Loop	`loop`	boolean	No	`false`	—	Conditional
Display	Initial item index	`initialIndex`	number	No	`0`	—	Conditional
Display	Edge	`edgePadding`	select	No	`bleed`	2 options	Conditional
Display	When there are no cards	`emptyBehavior`	select	No	`hide`	2 options	—
Display	Empty message	`emptyMessage`	text	No	—	—	Conditional
Display	Analytics	`analyticsEnabled`	boolean	No	`true`	—	—
Display	Style options	`classes`	multiselect	No	—	3 options	—

5.2 Card (`card`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Image	`image`	reference	No	—	—	—
Content	Image alt text	`imageAlt`	text	No	—	—	—
Content	Eyebrow	`eyebrow`	text	No	—	—	—
Content	Title	`title`	text	No	—	—	—
Content	Description	`text`	richtext	No	`""`	—	—
Content	Label	`subtitle`	text	No	—	—	—
Content	Tags	`tags`	aem-tag	No	—	—	—
Content	Date	`date`	date-time	No	—	—	—
Content	Content type	`contentLabel`	text	No	—	—	—
Content	Reading time	`readingTime`	text	No	—	—	—

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Link	Link	`link`	aem-content	No	—	—	—
Link	Link text	`linkText`	text	No	—	—	—
Link	Call to action label	`ctaLabel`	text	No	—	—	—
Link	Open in a new tab	`newTab`	select	No	`""`	3 options	—
Display	Card style	`kind`	select	No	`""`	5 options	—
Display	Card colour	`theme`	select	No	`""`	9 options	—
Display	Analytics label	`analyticsLabel`	text	No	—	—	—
Display	Accessibility label	`ariaLabel`	text	No	—	—	—

"Required in model" is the model's `required` flag; whether a field is required in practice is stated in the authoring guide where it is known.

6. Dependencies

Import
`../../utils/card-list-config.js`
`../../utils/dates.js`
`../../decorators/card-list-decorator.js`
`../../utils/cells.js`
`../../utils/environment.js`
`../../utils/placeholders.js`
`../../utils/properties.js`
`../../utils/tags.js`
`../../utils/taxonomy.js`
`./cards-data.js`

7. For Developers

```
blocks/cards/
├── README.md
├── _cards.json
├── cards-data.js
├── cards.css
└── cards.js
```

Decorator/Utility	Purpose	Execution or dependency point
`utils/card-list-config.js`	`normalizePresentation` (the card list settings of List and Cards, with defaults and validation), `PRESENTATION_PROPERTIES`, `kindOf` and `themeOf` (the card's Card style and Card colour: item, list, page metadata, autom	`cards-data.js`: { CARD_KINDS, CARD_THEMES, isExternal, isSafeUrl, isYes, text, toISODate, }
`utils/dates.js`	`formatDate` (DD-MM-YYYY, MM-DD-YYYY, MMM d, yyyy, DD/MM/YY, DD MMM YY, 4th September 2026; no time zone shift)	`cards-data.js`: { formatDate }

Decorator/Utility	Purpose	Execution or dependency point
decorators/card-list-decorator.js	buildCardListHeader, renderCardList, renderCard, renderRow, renderFeature, applyCardListLayout, buildCardListSkeleton (the List's skeleton in its own layout), renderCardListEmpty, cardListId, `cardHeadi	cards.js: { applyCardListLayout, buildCardListHeader, cardListId, renderCardList, renderCardListEmpty, renderEditorItems, trackCardList, }
utils/card-list-config.js	normalizePresentation (the card list settings of List and Cards, with defaults and validation), PRESENTATION_PROPERTIES, kindOf and themeOf (the card's Card style and Card colour: item, list, page metadata, autom	cards.js: { HEADINGS, PRESENTATION_PROPERTIES, normalizePresentation, pickValue, }
utils/cells.js	itemRows, KEY_VALUE_ITEM, FIELD_ROW_ITEM (the item rows of a block: by model in the editor, by their cells on the page), withoutTabCells (drops the empty cell AEM renders for each dialog tab of an item model), `m	cards.js: { matchCells, withoutTabCells }
utils/environment.js	prefersReducedMotion, scrollBehavior, isEditing, isInEditor, isAuthor (the page is rendered by AEM author, which has no query index)	cards.js: { isEditing }
utils/placeholders.js	getPlaceholders, loadPlaceholders, fillPlaceholder (UI text from the locale's placeholders, with English fallbacks)	cards.js: { getPlaceholders }
utils/properties.js	readProperties, textOf, isOn, isOnByDefault (a toggle that defaults to on), optionOf, normalizeKey, textSource (key-value block properties)	cards.js: { normalizeKey, textOf }
utils/tags.js	readTagValues, parseTag, readTags (tag IDs; ID-derived labels as the taxonomy fallback)	cards.js: { readTagValues }
utils/taxonomy.js	loadTaxonomy, resolveTags (tag labels from the locale's taxonomy)	cards.js: { resolveTags }

- **JavaScript (`cards.js`):** reads the block (`readBlock`), normalizes the settings (`normalizePresentation` with `idPrefix: 'cards'`, plus `emptyBehavior`), builds the header (`buildCardListHeader`), the cards (`buildManualCards`), resolves tag labels (`resolveTags`), applies the layout classes (`applyCardListLayout`, plus `source-manual`), renders (`renderCardList` or `renderCardListEmpty`), adds editor placeholders (`renderEditorItems`) and the data layer state (`trackCardList`).
- **Data (`cards-data.js`, no DOM):** `CARD_LAYOUT` (the Card cells in model order with `null` for the three dialog tab cells), `itemToCard` (one card; `null` for no content or an unsafe link), `buildManualCards` (authored order, ids renumbered after dropped cards).
- **CSS (`cards.css`):** only imports `styles/card-list.css` (the List's presentation).
- **Input DOM:** key-value rows (`key | value`, keys compared by `normalizeKey` against `PRESENTATION_PROPERTIES`); Card rows (in the editor `data-aue-model="card"`; on the page any row that isn't a setting row). A Card row with more than two cells is matched to the 16 fields with `withoutTabCells` and `matchCells` (by order and content: an image, a `YYYY-MM-DD` date, a link, `true` / `false`, a known style or colour, single-line text), so fields AEM leaves out don't shift the others; a row with two cells or fewer is a legacy card (image + rich text, `readLegacyText`).
- **Rendered DOM:** block classes `card-list`, `{bp}-{grid,scroll}`, `{bp}-controls`, `{bp}-indicator`, `snap`, `snap-{start,center}`, `hide-scrollbar`, `no-swipe`, `edge-bleed`, `equal-height`, `stretch`, `feature-first`, `display-teaser` or `links`, `style-{cardStyle}`, `ratio-{ratio}`, `panel-{height}`, style options, `source-manual`; custom properties `--list-{columns,gap,visible,card-width}-{bp}`; children `div.list-header`, optional `article.list-feature`, `div.list-scroller[role=region]` > `ul.list-items` > `li.list-item[data-card-id][data-cmp-id][data-cmp-position]` > `article.list-card.list-card-{kind}` (or `article.list-row`), `div.list-nav` (indicator, previous / next). Data attributes: `data-cards-decorated`, `data-analytics` (`custom` / `off`), `data-overflow` on the scroller.
- **Decoration flow:** guarded by `data-cards-decorated` (a second call does nothing); awaits the placeholders and, when cards have tags, the taxonomy.

- **Events and async:** the scroller (`createScroller`) listens to scroll, resize, keydown and pointer events; no custom events are dispatched.
- **Error handling:** invalid settings fall back to defaults; cards without content or with unsafe links are dropped; no cards: hidden (empty and `hidden` on the block and its wrapper) or the empty message; in the editor always the hint.
- **Exports:** `cards.js` default `decorate`; `cards-data.js` `CARD_LAYOUT`, `itemToCard`, `buildManualCards`.
- **Parent / child contracts:** Card rows keep their instrumentation on the rendered `li` (`moveInstrumentation`) and the rich text on its body; rows not rendered as cards become editor placeholders.
- **Data-layer hooks:** `data-cmp-id`, `data-cmp-position`, `data-cmp-interaction`, `data-analytics-label` on card links and articles.

8. Extending the Block

1. **A dialog field or value:** shared presentation fields go in `models/_card-list.json` (they apply to List and Cards); Card fields in `blocks/cards/_cards.json`. Run `npm run build:json` and `npm run lint`.
2. **Map it to rendering:** settings in `utils/card-list-config.js` `normalizePresentation` (add the normalized key to `PRESENTATION_PROPERTIES`, or the row is read as a Card); Card values in `cards.js` `readCard` and `cards-data.js` `itemToCard`; markup in `decorators/card-list-decorator.js`; styles in `styles/card-list.css` (scoped to `.card-list`, mobile first, `min-width` 600 / 900 / 1200px, rem).
3. **Card cells:** a new Card field changes the row's cells: update `CARD_LAYOUT` (fields in model order, `null` per dialog tab), `CARD_MATCHERS` and `test/cards/cards-data.test.mjs`.
4. **Delivery rules:** don't end a field name with Text, Title, Type, Alt or MimeType unless it should collapse; tab cells arrive empty; Additional tab values arrive as `classes_*` block classes.
5. **Responsive, accessibility, data layer:** keep one link per card, region naming and focus rings; describe new clicks with `data-cmp-interaction` instead of pushing events.
6. **Tests:** `npm test` (`test/cards/cards-data.test.mjs`, and the List tests for shared code).
7. **Backward compatibility:** keep reading removed fields (Component ID, Touch swipe) and legacy two-cell cards; changes to the shared files affect the List.
8. **Docs:** update the block library pages (`block-library/cards/…`), this README (`npm run build:docs` regenerates its generated sections) and the List README; run `npm run build:modules` after changing imports.

9. Responsive Behavior

Viewport	Layout	Image behavior	Interaction behavior
Mobile (below 600px)	Mobile display: grid with Mobile columns (default 1, gap 16) or a rail with Mobile items visible (default 1.2) / card width; header stacked; smallest type scale	Image ratio kept; authored picture cloned	Mobile scroll indicator on by default, buttons off; touch scroll native
Tablet (600–1199px)	Tablet display: grid (default 4 columns, gap 24) or rail (default 3.3 visible); larger type; row dividers with Dividers between rows	Feature min-height 27.1875rem	Tablet buttons / indicator off by default
900px and wider	Card padding increases (`--space-600`)	—	—
Desktop (1200px and wider)	Desktop display: grid (default 4 columns, gap 24) or rail (default 4.2 visible); largest type scale	Feature min-height 35.625rem	Desktop buttons / indicator off by default; mouse drag on rails

Buttons and indicator are hidden whenever the cards fit (no overflow). Edge "Cards run to the edge" only differs from "Inside the content width" in sections with a side gutter.

10. Accessibility

- **Semantic HTML:** the cards are a `<ul>` / `<li>` list inside a `div.list-scroller` with `role="region"`, named by the title (`aria-labelledby`), else the Accessible list label, else "Cards" (placeholder `cards`); each card is an `<article>`; dates are `<time datetime>`.
- **Headings:** the block heading at the chosen level; card titles one level below (at most `h6`); a card without a title has no heading.
- **Keyboard and focus:** one link per card (the call to action when shown, else the title) whose overlay covers the card; inline links of the description stay operable above it; visible focus rings (inside the edge of panel, report and statistic cards). The scroller is focusable only while it overflows; the previous / next buttons use `aria-disabled` (not `disabled`) at the ends so the focus stays.
- **Accessible names:** call to action "{label}: {title}" unless an Accessibility label is set; previous / next buttons "Previous items" / "Next items" with `aria-controls`; new-tab links get the hidden text " (opens in a new tab)".
- **Alt text:** the Image alt text; empty means decorative.
- **Announcements:** none (no live regions).
- **Reduced motion:** scrolling is instant for `prefers-reduced-motion` (always honoured); the row arrow transition is off; the content animation doesn't play.
- **Contrast:** the theme contract turns image card text light on dark Section / Columns Container backgrounds; panel, report and statistic cards keep their Card colour. The handover notes check points for Brand blue and Image background cards.
- **Empty / loading states:** no skeleton (authored content renders at once); an empty block is hidden or shows a message as text.
- **Reflow:** single column below 600px by default; long numbers scale down in narrow statistic cards.

Author responsibilities: a title for every card and the block (or an Accessible list label); meaningful alt text; heading type that fits the outline; an Accessibility label mentioning the new tab for new-tab calls to action; not relying on colour alone.

Developer responsibilities: keep the one-link-per-card pattern and the region naming in `decorators/card-list-decorator.js`; keep focus rings visible in `styles/card-list.css`; keep the scroller's `aria-disabled` behaviour (`decorators/scroller-decorator.js`); UI text only from placeholders.

Key	Expected behavior
Tab / Shift+Tab	Moves through the header link, the card links, inline description links, the scroller (while it overflows) and the previous / next buttons
Enter	Follows the focused link; activates a focused previous / next button
Space	Activates a focused previous / next button
Arrow Right / Arrow Left	On the focused scroller (Keyboard scrolling on): next / previous step
Home / End	On the focused scroller: first card / end of the row

11. Data Layer and Analytics

- **Component:** with Analytics on (default), `trackCardList` marks the block `data-analytics="custom"` and pushes the block (`@type: tcs-corp/components/cards`, `parentId` the page, `dc:title` the heading or Accessible list label, `xdm:variant` the Card style or `links`) and each shown card (`@type: tcs-corp/components/cards/item`, `parentId` the block, `dc:title`, `xdm:linkURL`, `xdm:position`, `repo:modifyDate` the card date).
- **Ids:** the stored Component ID, else the Anchor ID (block id), else `cards-1`, `cards-2`, ...; cards `{id}-item-{n}` in the shown order (they change when cards are reordered).
- **Interactions:** clicks are captured centrally by `decorators/data-layer-decorator.js` from `data-cmp-id`, `data-cmp-position` (on each `li` and the feature card) and `data-cmp-interaction` (`card`, `cta`, `previous`, `next`). A click ending a mouse drag and clicks on `aria-disabled` buttons aren't pushed. Cards makes no `pushComponentEvent` call of its own.
- **Analytics off:** `data-analytics="off"`: no state and no events for the block.

- **Path:** Adobe Client Data Layer → Adobe Data Collection (Tags) → AEP Web SDK → Edge Network → Analytics / CJA is the intended path, but Tags and the Web SDK aren't configured yet (blocked: measurement plan). No consent mechanism is implemented.

Trigger	Data-layer event	Required properties	Analytics behavior
Block decorated (Analytics on)	`component` state	block `@type`, `parentId`, `dc:title`, `xdm:variant`; per card `@type`, `parentId`, `dc:title`, `xdm:linkURL`, `xdm:position`	Impression state for the block and its cards
Card link or call to action	`cmp:click`	`path: component.{card id}`, `interaction: card`, `dc:title`, `xdm:linkURL`, `xdm:linkType` (`other`, `exit`, `download`), `position`, `breakpoint`	Card click with its position
Link inside a card description	`cmp:click`	`path: component.{card id}`, `interaction: link`, `position`	Inline link click
Header CTA	`cmp:click`	`path: component.{block id}`, `interaction: cta`, `dc:title`, `xdm:linkURL`	CTA click
Previous / Next	`cmp:click`	`path: component.{block id}`, `interaction: previous` / `next`, `dc:title` (button name), `breakpoint`	Scroller navigation

12. Performance Considerations

- **Initialization:** decorated with its section (eager only in the first section); waits for the placeholders and, when cards have tags, the taxonomy (both cached per page).
- **Images:** the authored `<picture>` is cloned; `loading` from Image loading (lazy by default); with Eager only the first image gets `fetchpriority="high"`; aspect ratios reserve space (no layout shift).
- **Dynamic imports / requests:** none beyond the block's own modules; no query index.
- **DOM size:** one `li` and `article` per card; no virtualization.
- **Listeners / observers:** the scroller adds scroll, resize (ResizeObserver), keydown and pointer listeners on the block; the motion decorator observes the block when an animation is set.
- **Layout shift risks:** the block rebuilds its children on decoration; hiding an empty block removes its wrapper from the layout.
- **High-cost configurations:** many cards with Eager loading; large rails with fixed card widths.

13. Security and Privacy

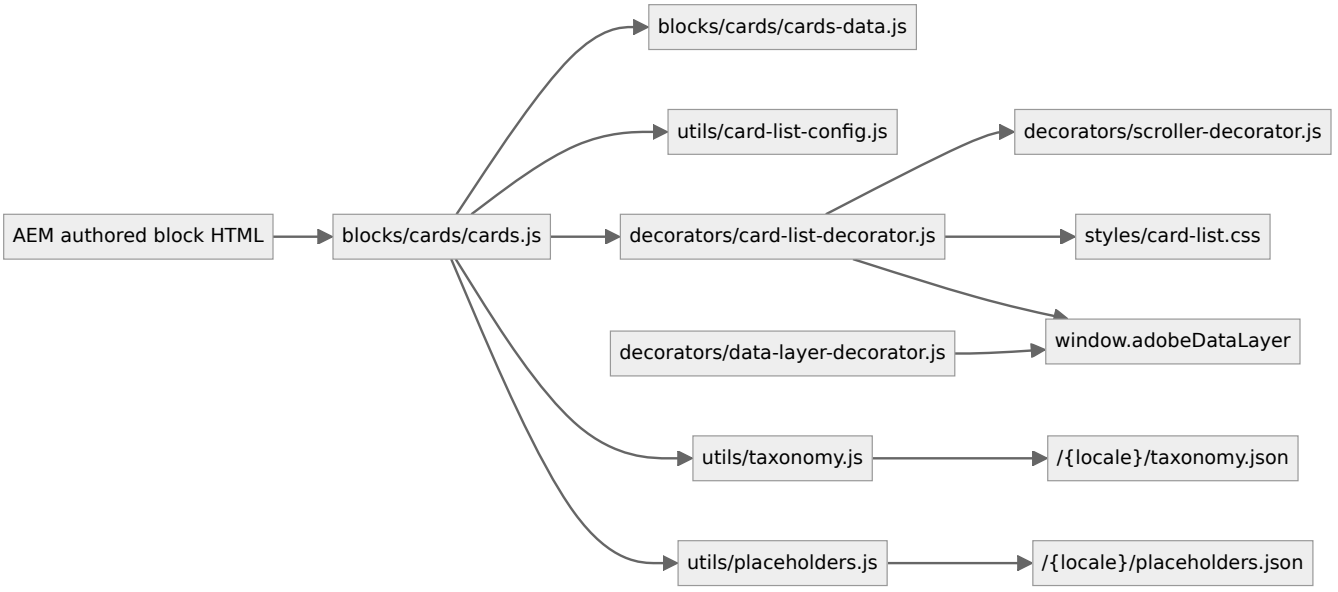
- Card links are validated with `isSafeUrl` (http, https, mailto, tel, paths, anchors and queries; any other scheme such as `javascript:` or `data:` drops the card).
- The header CTA is the link AEM delivers; it isn't validated by the block.
- Text values are set with `textContent`; the rich text description is the AEM-delivered markup moved as is.
- New-tab links get `rel="noopener noreferrer"`; links to other origins are marked `data-external="true"`.
- No external content, query string handling or personal data; the data layer holds titles, URLs and positions only.

14. For Architects

- **Role:** the manual counterpart of the List: same presentation, authored content.
- **Content model:** key-value container (`cards`) with item children (`card`); the settings are shared with the List.
- **Rendering and data source:** client-side decoration of server-delivered authored HTML; only the placeholders and (with tags) the taxonomy are fetched.
- **Dependencies:** `decorators/card-list-decorator.js` (with the scroller, button, icon button, link, heading, responsive image, skeleton and data layer decorators), `utils/card-list-config.js`,

`utils/cells.js`, `utils/taxonomy.js`, `utils/placeholders.js`, `utils/analytics.js`.

- **Data layer / Web SDK / Target:** Adobe Client Data Layer state and clicks; Tags, Web SDK and Target aren't configured.
- **Consent and identity:** none implemented; no personal data in the data layer.
- **Caching:** the page HTML (with the cards) is cached by the Edge Delivery CDN and refreshed on publish; placeholders and taxonomy are fetched once per page and cached in memory.
- **Performance:** no data requests beyond placeholders and taxonomy; CSS-driven layout; lazy images by default.
- **Security:** unsafe card links are dropped; text set with `textContent`.
- **Scalability:** no maximum number of cards; DOM size grows with the cards.



15. Troubleshooting

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
A Card is missing	No title, description or image, or an unsafe link	Check the Card's fields and Link scheme	Add content; use the page picker or an `https://` URL	The card shows after reload
The whole block is missing	No card content with Hide the block; Hide block on; script load error	Edit mode hint; Additional tab; console `failed to load block cards`	Add content or Show a message; turn Hide block off; escalate load errors	Block visible on preview and live
Tags show ID-like labels	Tag missing in the locale taxonomy or taxonomy not published	Search the tag ID in `/{locale}/taxonomy.json`	Add / translate the tag, publish the taxonomy	Correct label in a new session
Text clipped in coloured cards	Portrait / Compact panel height	Check Panel height and text length at narrow widths	Shorter text or Panel height Content	All text visible at all widths
No previous / next or indicator	Breakpoint in grid mode, setting off, or no overflow	Check the breakpoint's fields; inspect `.list-scroller[data-overflow]`	Turn the controls on for the scrolling breakpoint; fewer items visible	Controls show and work
Settings have no effect	Block authored before the redesign (no block model)	Check `.plain.html` for key-value rows	Replace with a new Cards block	Block follows the settings
Date or reading time missing	Show date / Show tags off; card kind without date (panel, statistic)	Check the switches and card kind	Turn on; use image cards for dates	Date shows in the Date format
Call to action missing	No Link, Link items off or Show call to action off	Check the Card Link and switches	Add the Link; turn the switches on	Button shows

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
No clicks in the data layer	Analytics off; the click ended a drag	Check `data-analytics`; inspect `window.adobeDataLayer`	Turn Analytics on	`cmp:click` entries appear
Card goes through a redirect	Authored link not updated after a page move	Check `/{locale}/redirects.json`	Update the Link and republish	Card opens the target directly

Evidence to collect: page URL (preview / live), viewport width, the block's `.plain.html` rows, the block's classes and `data-analytics`, console messages, the relevant `window.adobeDataLayer` entries. No credentials or tokens.

16. Known limitations of the block

Limitation	Impact	Workaround	Planned correction
Cards blocks created before the redesign have no block model	Their settings can't be edited	Replace with a new Cards block	Migration (Phase 0, `docs/proposals/cards-card-refactoring.md`)
Those blocks render in the new design, not the original bordered grid	Visual change on existing pages	Replace or accept	Classic presentation (Phase 0)
In legacy cards the title stays inside Description in the Card dialog	Authors edit the title in the rich text	Format the title as a heading	Move titles into Title when editing (proposal)
No author lockup on the feature card	Feature without author	Use the List for page-based features	Not recorded
Internal links aren't resolved through the redirects sheets	Moved pages go through a redirect	Update links after page moves	Not recorded
Card text fields are edited in the properties panel; only the description in place	Slower editing	—	Not recorded
The new-tab hint isn't announced on calls to action without an Accessibility label (the generated `aria-label` replaces it)	Screen reader users aren't told	Mention the new tab in the Accessibility label	Not recorded
Month abbreviations are English in every locale	Non-English dates in English	Use a numeric Date format	Not recorded
Card ids follow the shown order	Reordering changes the data-layer ids (D-A-23)	—	Deferred (audit)
Edge "Cards run to the edge" looks like "Inside the content width" in full-width sections (D-A-9)	No visible difference	—	Deferred (audit)
A stored Touch swipe off with no controls blocks touch scrolling (D-A-10)	Cards unreachable on touch	Turn on previous / next buttons	Deferred (audit)
No carousel autoplay or dots, masonry, filtering, sorting or search	—	Filter List for filterable listings	Planned (proposal phases 1–6)
Card date may show the previous day for some stored date-times (D-A-26, suspected)	Wrong date	Check the published date	Blocked (to be verified)

17. Change history of the block

Version	Change	Source	Date
Not recorded	README generated from verified implementation and approved handover documentation	Repository review	Not recorded

Version	Change	Source	Date
Handover 1.1	Cards and Card analysis: blocks made before the redesign have no block model and changed appearance (refactoring proposal)	docs/handover revision history	14 October 2026
Handover 2.7	The Display as teaser block class is `display-teaser` (was `teaser`, also selected by the Teaser styles), F-83	docs/handover revision history	17 October 2026
Handover 2.8	Dialog tabs Content / Display / Additional; an item without a card is a placeholder in the editor (`cardEditorHint`), F-84, F-85	docs/handover revision history	17 October 2026
Handover 2.18	Additional tab Content animation (repeated items staggered)	docs/handover revision history	18 October 2026
Handover 2.27	Block ecosystem audit: header CTA style applied, focus ring inside panel / report / statistic cards, Card colour only on those cards, previous / next keep the focus, Component ID and Reduced motion support removed from the dialog	docs/handover revision history	19 October 2026
Handover 2.34	Theme contract: List / Cards image cards light on dark backgrounds	docs/handover revision history	20 October 2026

18. Reference pages

Kind	Page	Live URL
Block library	/design-block-master/en/block-library/cards	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/cards
Block library	/design-block-master/en/block-library/cards/card-colours	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/cards/card-colours
Block library	/design-block-master/en/block-library/cards/card-styles	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/cards/card-styles
Block library	/design-block-master/en/block-library/cards/grid-and-horizontal-scroll	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/cards/grid-and-horizontal-scroll
Block library	/design-block-master/en/block-library/cards/heights-links-legacy	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/cards/heights-links-legacy
Block library	/design-block-master/en/block-library/cards/images-panels-style-options	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/cards/images-panels-style-options
Block library	/design-block-master/en/block-library/cards/per-card-styles-and-colours	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/cards/per-card-styles-and-colours
Curated design page	/design-block-master/en/curated-design-pages/industry-landing-page	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/curated-design-pages/industry-landing-page
Curated design page	/design-block-master/en/curated-design-pages/product	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/curated-design-pages/product
Curated design page	/design-block-master/en/curated-design-pages/services	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/curated-design-pages/services
Curated design page	/design-block-master/en/curated-design-pages/who-we-are	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/curated-design-pages/who-we-are

19. Coverage of the required topics

Topic	Where
Purpose	Overview

Topic	Where
Source files	generated: Files
Model, definition, filters, children	generated: Components
Dependencies	generated: Dependencies
Authored input / rendered output / DOM transformation	For Developers
Configuration fields, defaults, required fields	generated: Fields
Variations	CARDS-VARIATIONS.md
Responsive behaviour	Responsive Behavior
Accessibility and keyboard	Accessibility
Analytics / data layer	Data Layer and Analytics
Performance	Performance Considerations
Security	Security and Privacy
Extending	Extending the Block
Known limitations	Known Limitations
Troubleshooting	Troubleshooting
Tests	CARDS-TEST-COVERAGE.md
SEO behaviour	Not recorded
Caching considerations	Not recorded (Edge Delivery default caching; no block-specific rule)
Ownership	Technical Lead / Block Owner (proposed)

20. Screenshots

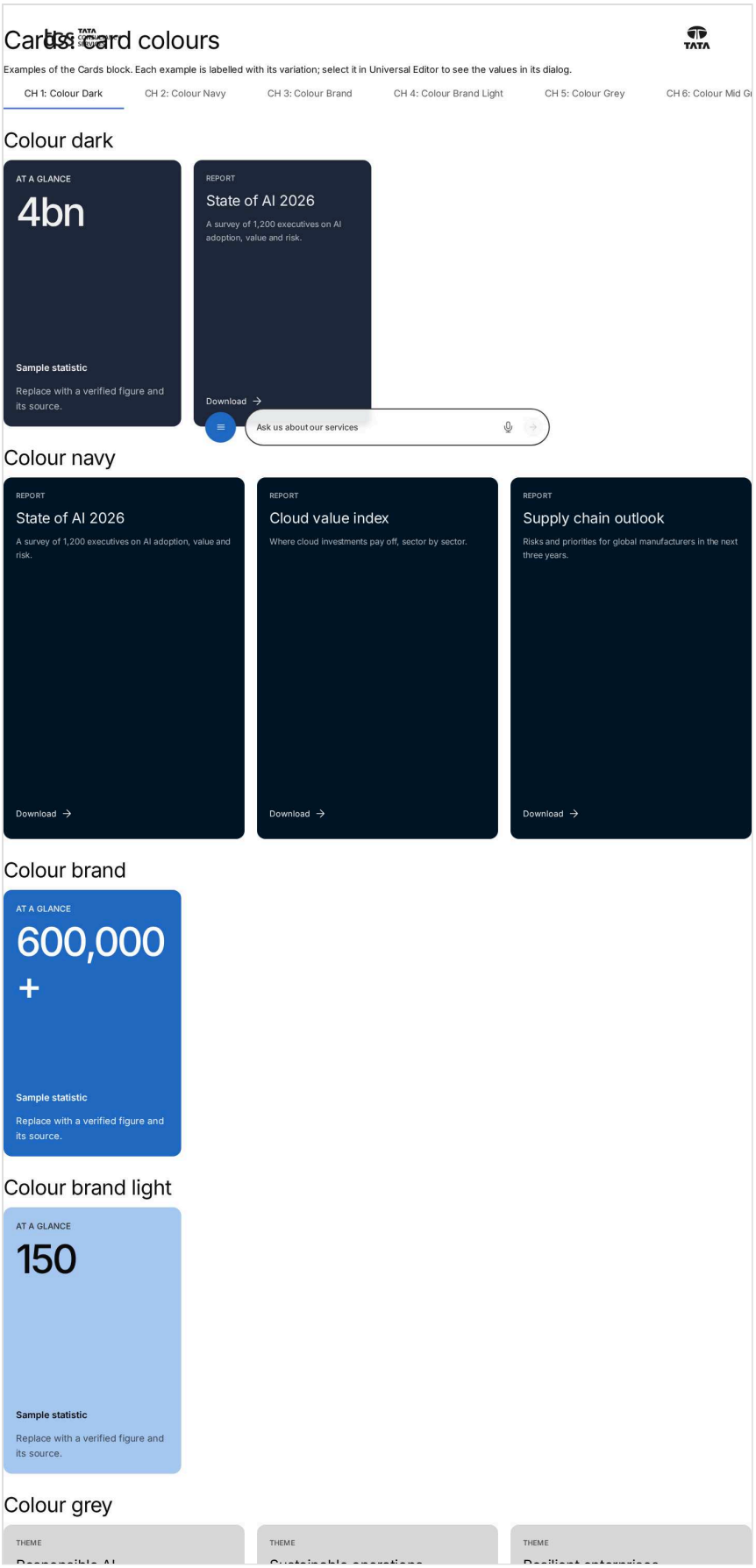


Figure: Cards block library page — desktop 1440 × 900 (/design-block-master/en/block-library/cards/card-colours, captured 2026-11-08).

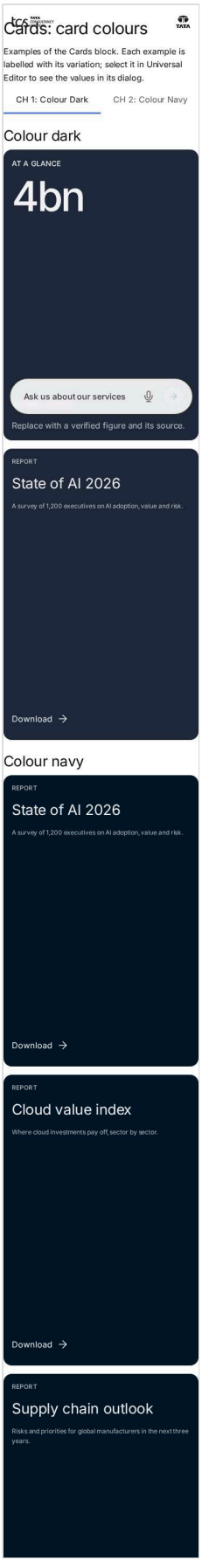


Figure: Cards block library page — mobile 375 × 812 (/design-block-master/en/block-library/cards/card-colours, captured 2026-11-08).

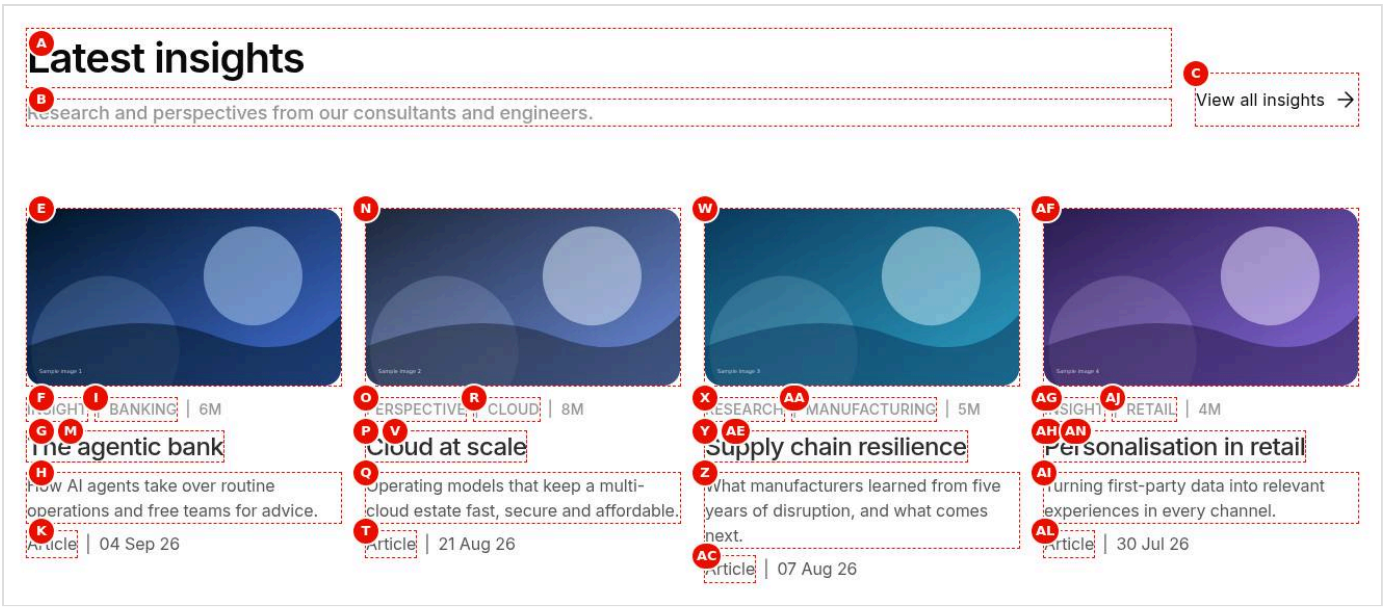


Figure: Desktop rendering — The Cards block at 1440 px with default settings: the header with the H2 "Latest insights" and its description on the left and the "View all insights" text link on the right, above four image cards in one row (photo-1.svg to photo-4.svg in 16:9 frames, kickers such as "Insight", the tag label and "6m", titles from "The agentic bank" to "Personalisation in retail", descriptions and meta rows "Article" with dates from "04 Sep 26"). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-desktop-view.jpg) (handover screenshot; rendering at the time it was taken).

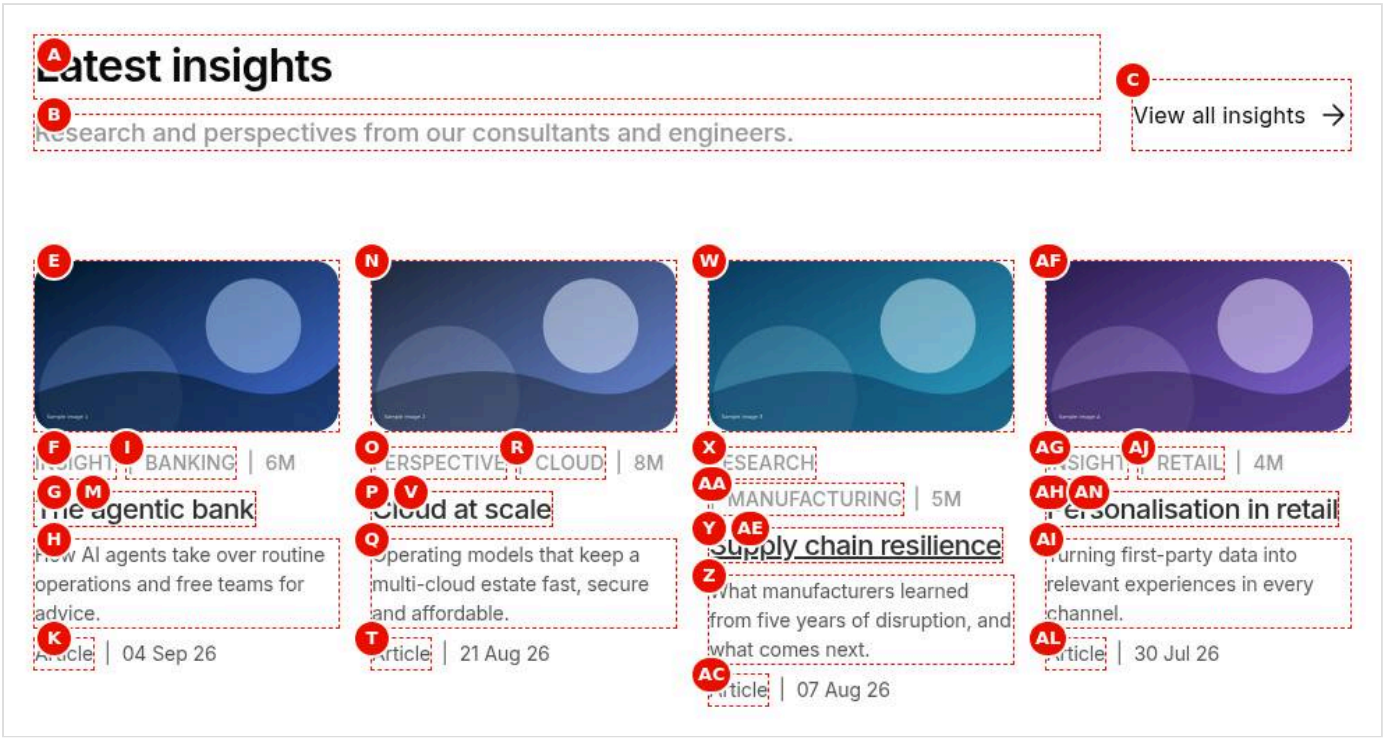


Figure: Tablet rendering — The same Latest insights block at 1024 px: the 2rem title with "View all insights" still on the right, and the four image cards in one row (Tablet columns 4 by default) with 1.25rem card titles and 0.875rem kickers and meta rows. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-tablet-view.jpg) (handover screenshot; rendering at the time it was taken).



Figure: Mobile rendering — The Latest insights block at 402 px: the header stacked (title, description, then "View all insights" below), and the four image cards one per row in a single column with a 16 px gap, 1.125rem titles and 0.75rem kicker and meta rows. (Sample content on the preview; callout letters as in

the Field Entry Table; file screenshots/cards-mobile-view.jpg) (handover screenshot; rendering at the time it was taken).

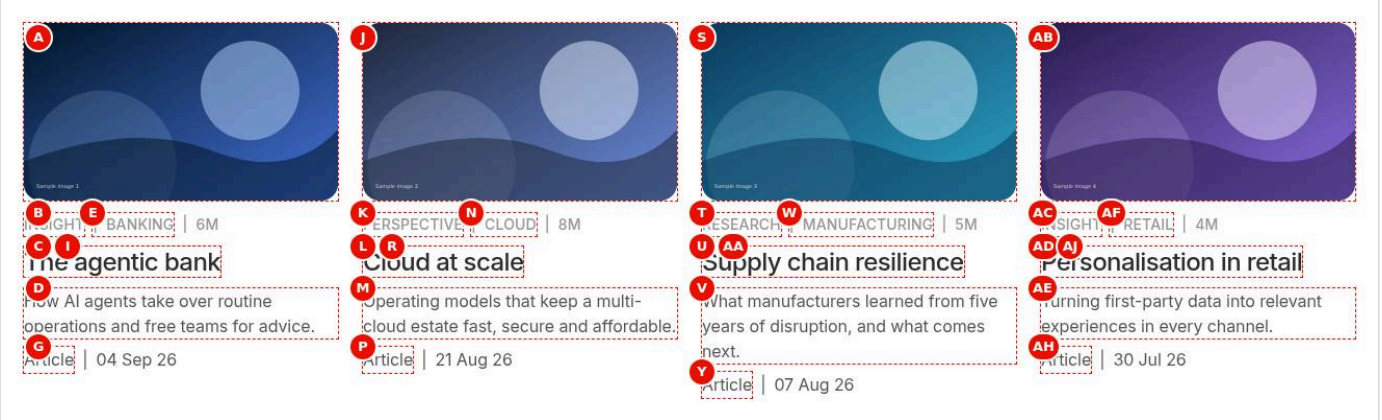


Figure: Image card style — Four image cards in a row at 1440 px (default Card style Image card, 4 columns): 16:9 rounded images, uppercase grey kickers (e.g. "Insight", the tag label, "6m"), the titles "The agentic bank", "Cloud at scale", "Supply chain resilience" and "Personalisation in retail", two-line descriptions and the meta row "Article" with the date (e.g. "04 Sep 26"). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-image-card.jpg) (handover screenshot; rendering at the time it was taken).

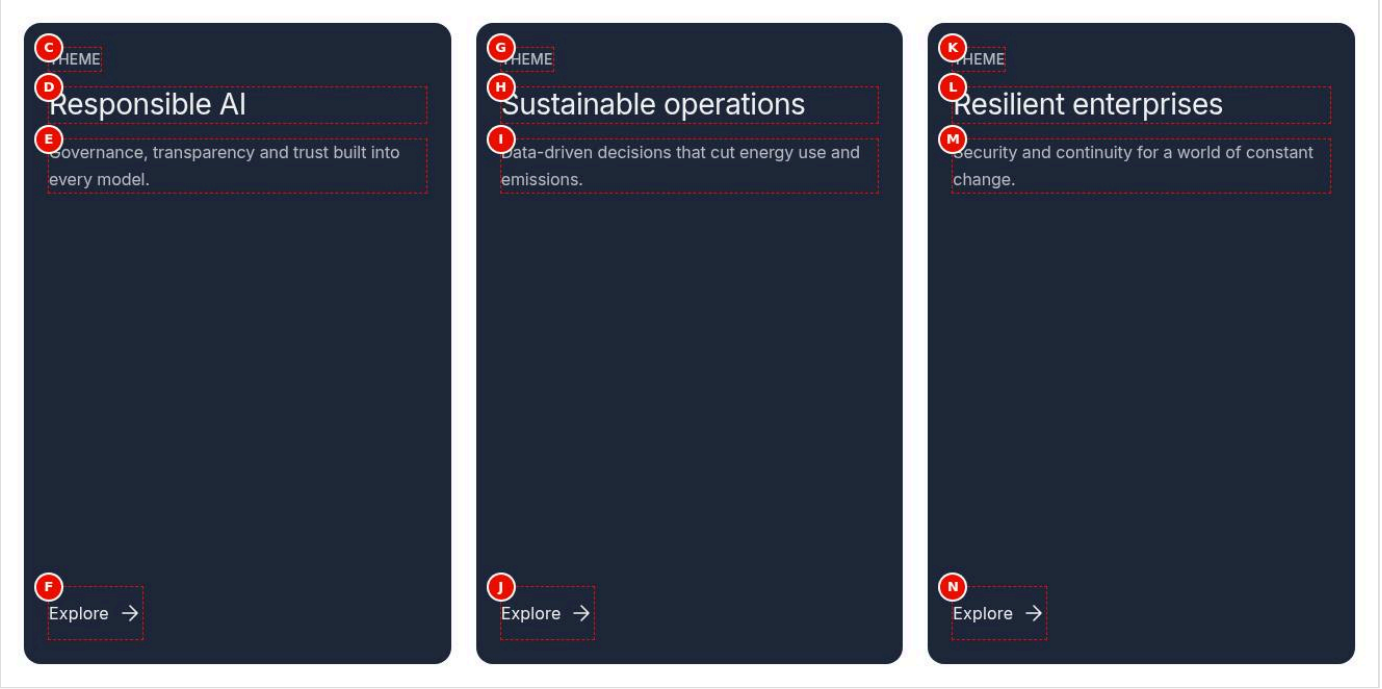


Figure: Panel card style — Three tall 2:3 dark navy-grey (#20293c) panels in a 3-column row at 1440 px, each with the light kicker "Theme", a large light title ("Responsible AI", "Sustainable operations", "Resilient enterprises"), a muted description and the "Explore" text link at the bottom. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-panel-card.jpg) (handover screenshot; rendering at the time it was taken).

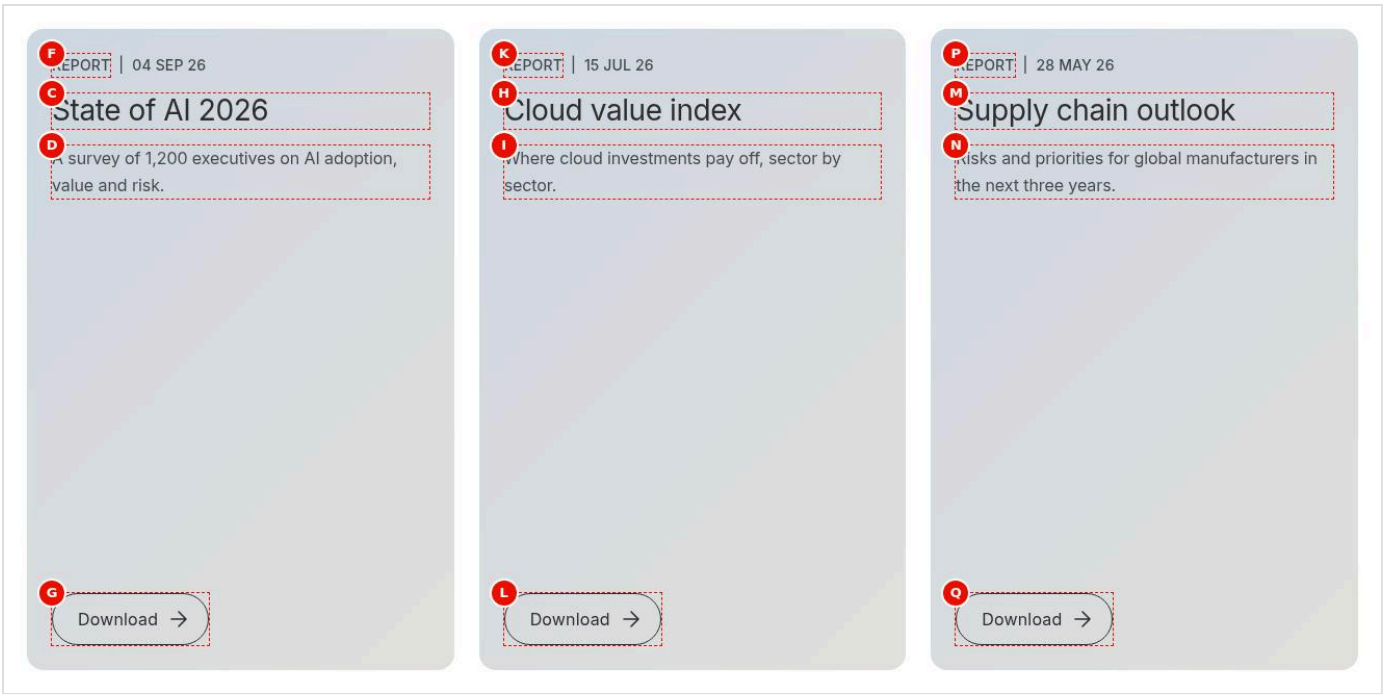


Figure: Report card style — Three tall 2:3 report cards in a 3-column row at 1440 px on the light blue-grey to warm grey gradient: kickers "Report" with the dates "04 Sep 26", "15 Jul 26" and "28 May 26", the dark titles "State of AI 2026", "Cloud value index" and "Supply chain outlook", descriptions and an outlined "Download" button at the bottom. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-report-card.jpg) (handover screenshot; rendering at the time it was taken).

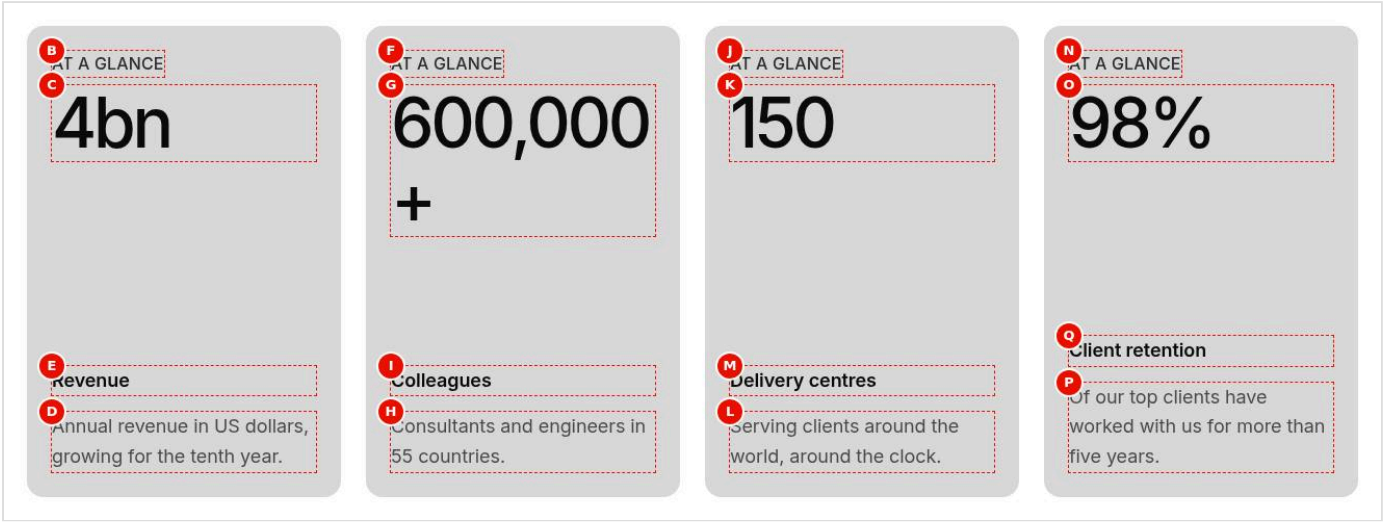


Figure: Statistic card style — Four light grey (#d9d9d9) 2:3 statistic cards in a row at 1440 px, each with "At a glance" at the top, a very large number ("4bn", "600,000+", "150", "98%") and at the bottom a bold label ("Revenue", "Colleagues", "Delivery centres", "Client retention") with a short description; no links or buttons. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-statistic-card.jpg) (handover screenshot; rendering at the time it was taken).

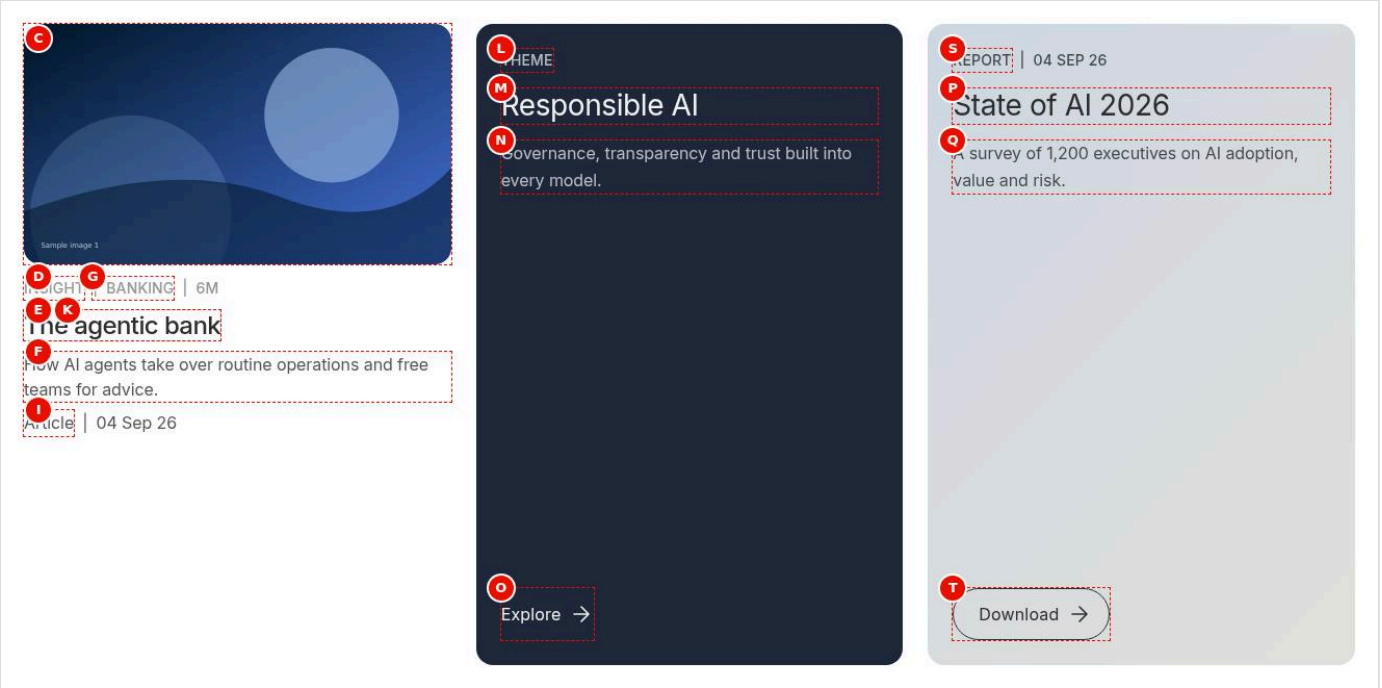


Figure: Mixed card style — A 3-column row at 1440 px with Card style Mixed: the image card "The agentic bank" (photo-1.svg), the dark #20293c panel "Responsible AI" (no image) with "Explore", and the gradient report card "State of AI 2026" (Content type Report) with the kicker "Report", "04 Sep 26" and a Download button. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-mixed.jpg) (handover screenshot; rendering at the time it was taken).

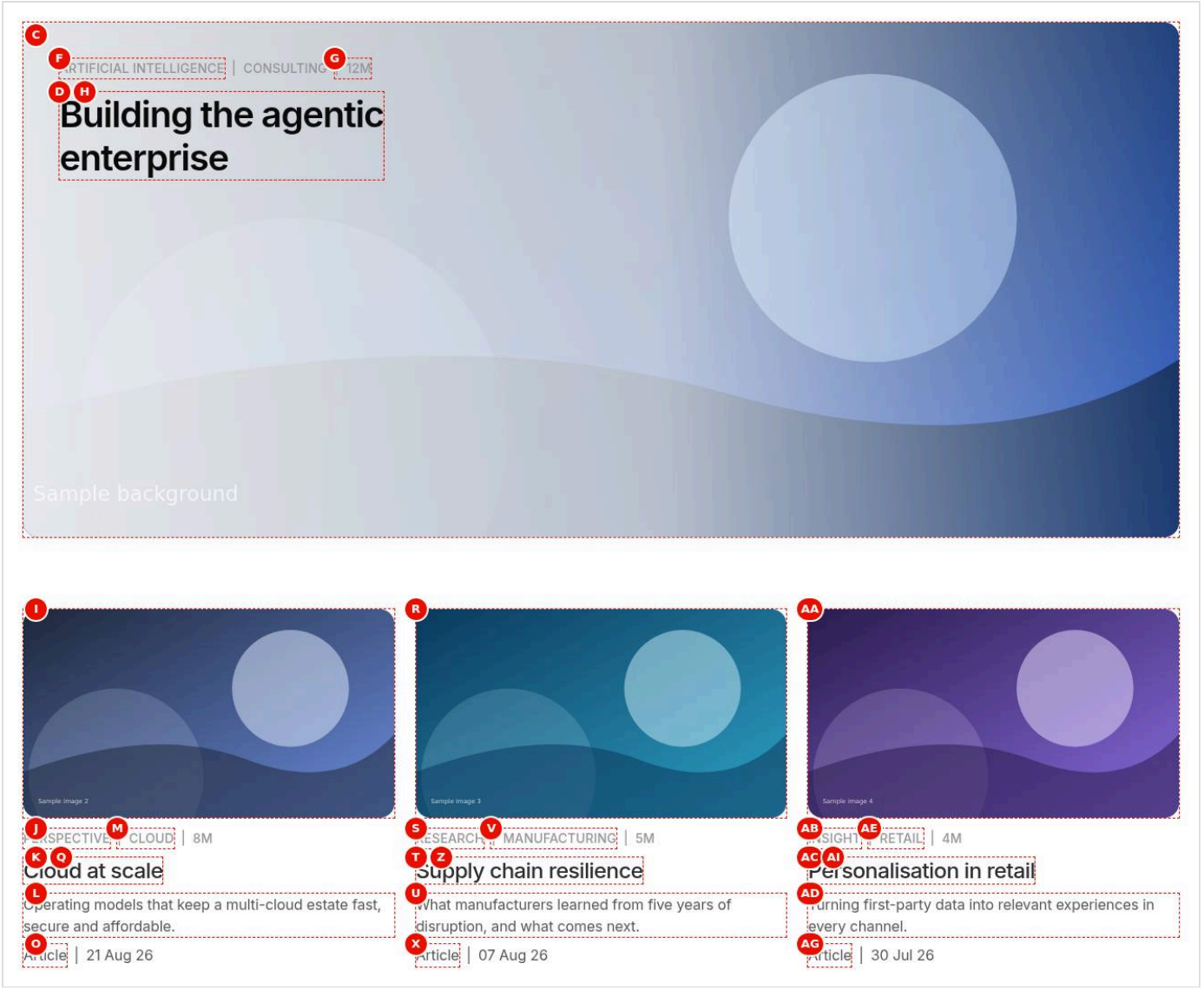


Figure: Feature card — At 1440 px a full-width feature card (about 35.6rem high) with wide.svg filling it under a white gradient from the left, the two tag labels and "12m" and the large title "Building the agentic enterprise" (no description); below it three image cards ("Cloud at scale", "Supply chain resilience", "Personalisation in retail") in a 3-column row. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-feature-card.jpg) (handover screenshot; rendering at the time it was taken).

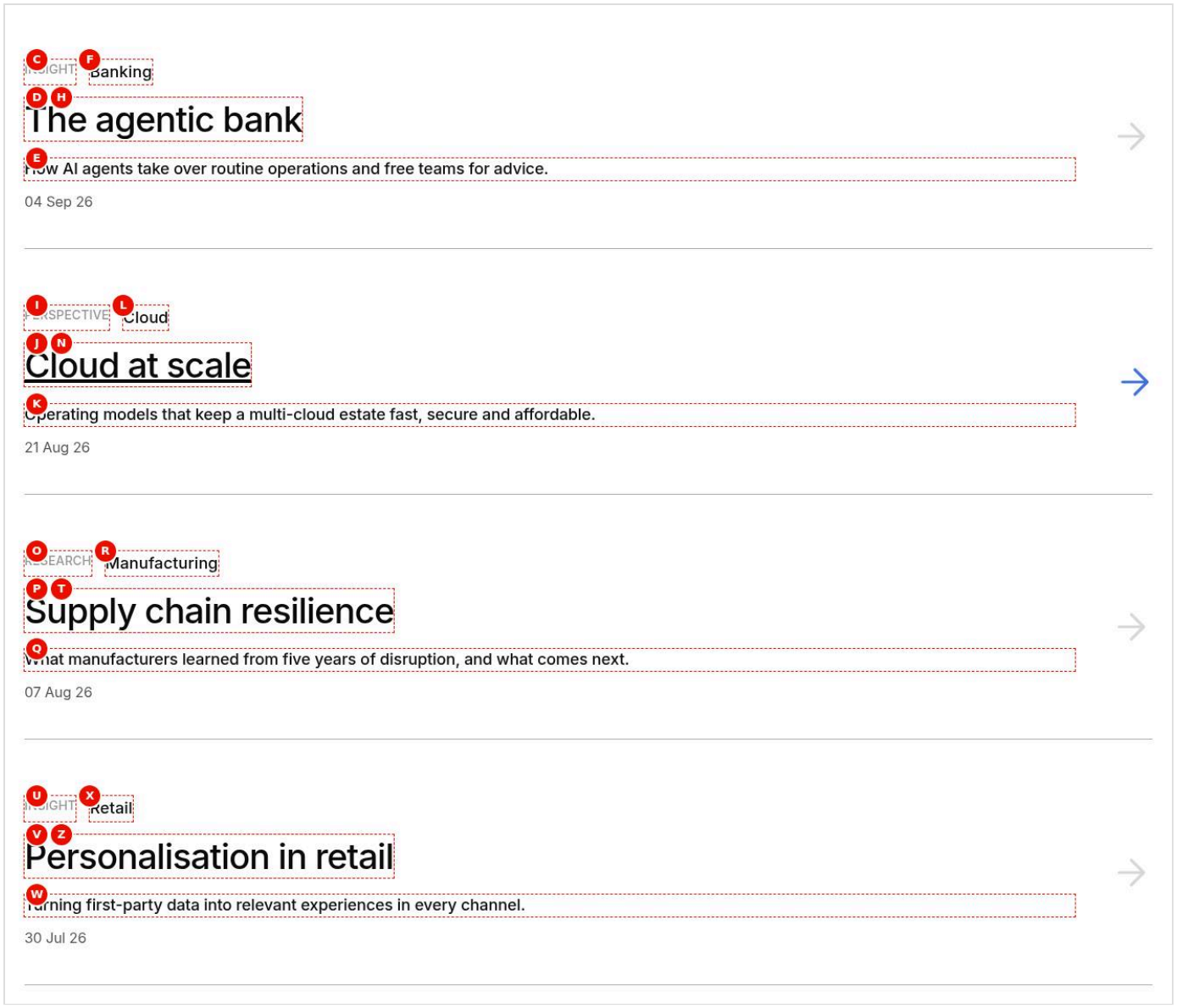


Figure: Rows of links — Four full-width rows of links at 1440 px, each with the uppercase eyebrow and first tag, a large title ("The agentic bank", "Cloud at scale", …), a description, the date and a grey arrow on the right, separated by thin grey lines (Dividers between rows); the pointer is on the second row, whose arrow is blue and moved. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-rows-of-links.jpg) (handover screenshot; rendering at the time it was taken).

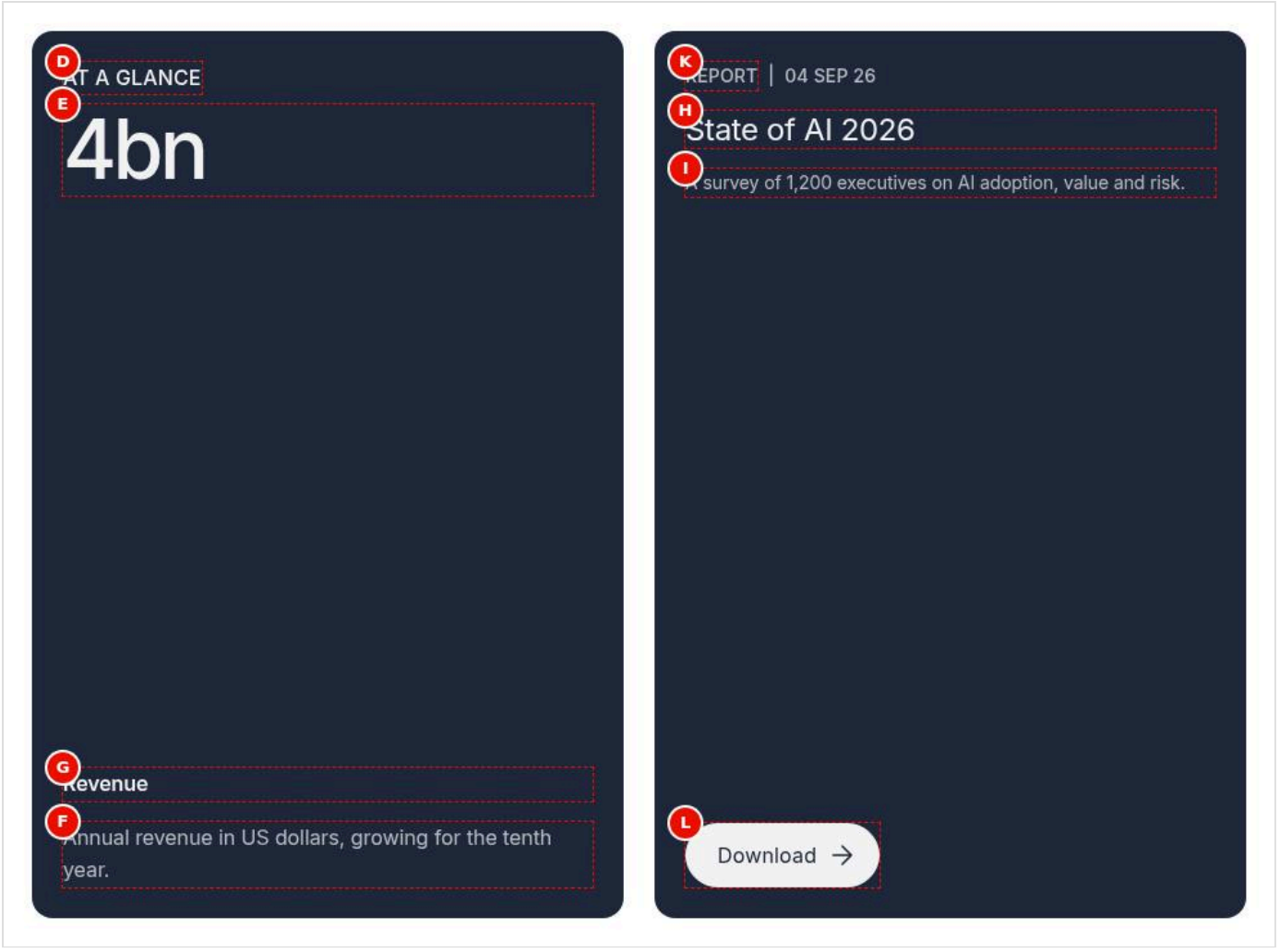


Figure: Dark card colour — At 1024 px two cards side by side (Tablet columns 2) on dark #20293c with light text: the statistic "4bn" with "At a glance", "Revenue" and its description, and the report card "State of AI 2026" with the kicker "Report", "04 Sep 26", the description and a light outlined Download button. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-colour-dark.jpg) (handover screenshot; rendering at the time it was taken).

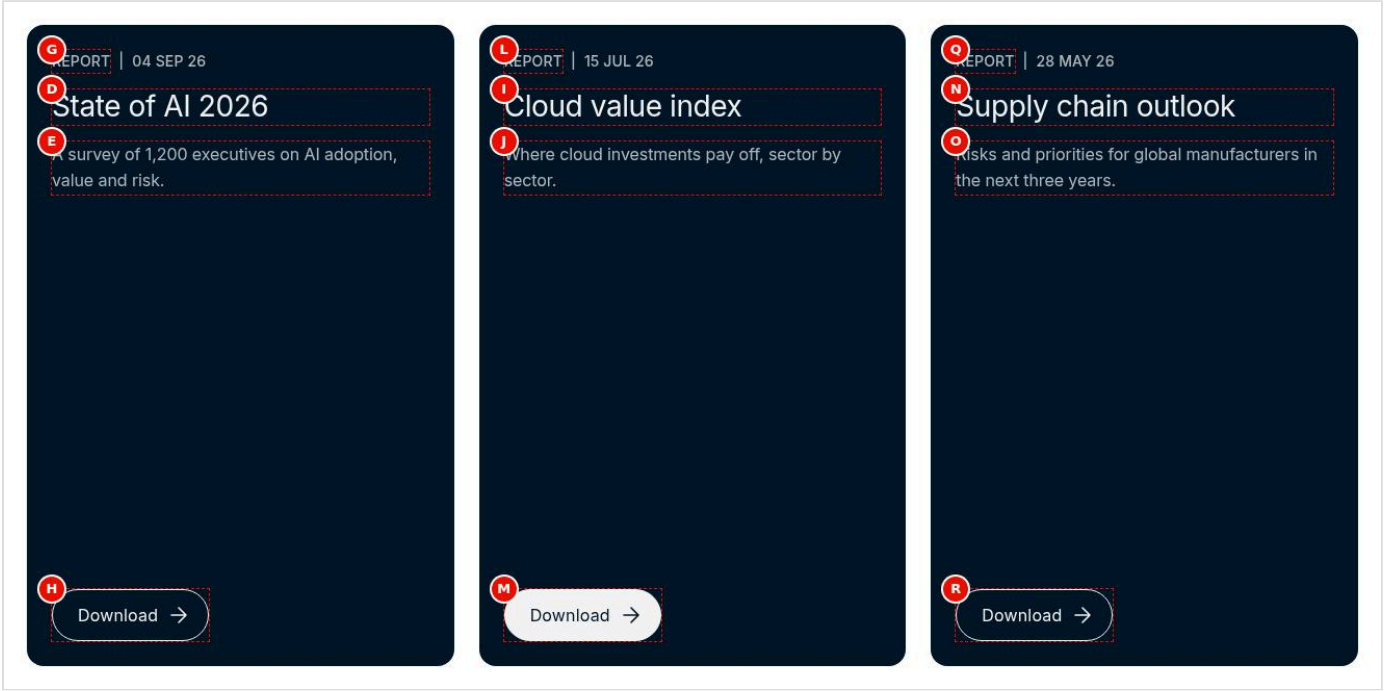


Figure: Navy card colour — Three 2:3 report cards in a 3-column row at 1440 px on near-black navy (#011627) with light text: the kicker "Report" with the date, the titles "State of AI 2026", "Cloud value index" and "Supply chain outlook", muted descriptions and a light outlined Download button at the bottom. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-colour-navy.jpg) (handover screenshot; rendering at the time it was taken).

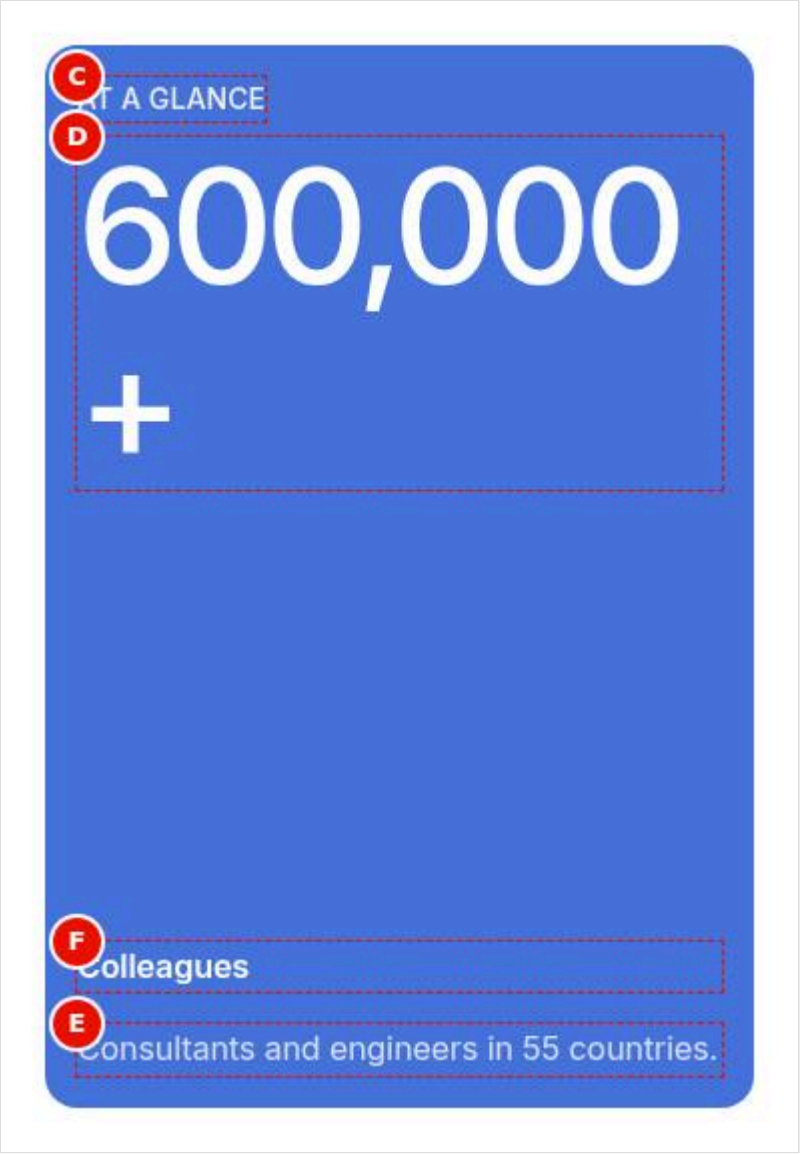


Figure: Brand blue card colour — At 402 px a single brand blue (#4673db) statistic card filling the column: "At a glance" at the top, the white number "600,000+", and at the bottom the label "Colleagues" with the description in 60 % white. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-colour-brand.jpg) (handover screenshot; rendering at the time it was taken).

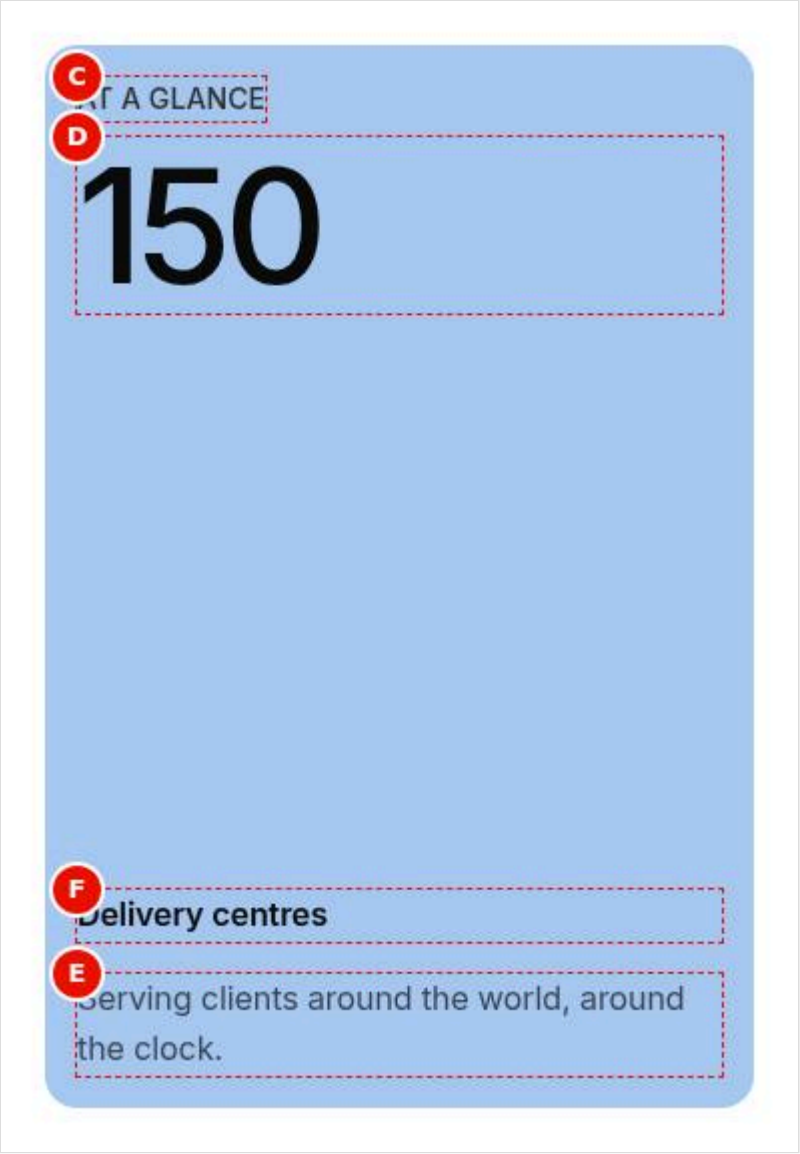


Figure: Light blue card colour — At 402 px a single light blue statistic card filling the column: "At a glance" at the top, the black number "150", and at the bottom the label "Delivery centres" with a dark description. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-colour-brand-light.jpg) (handover screenshot; rendering at the time it was taken).

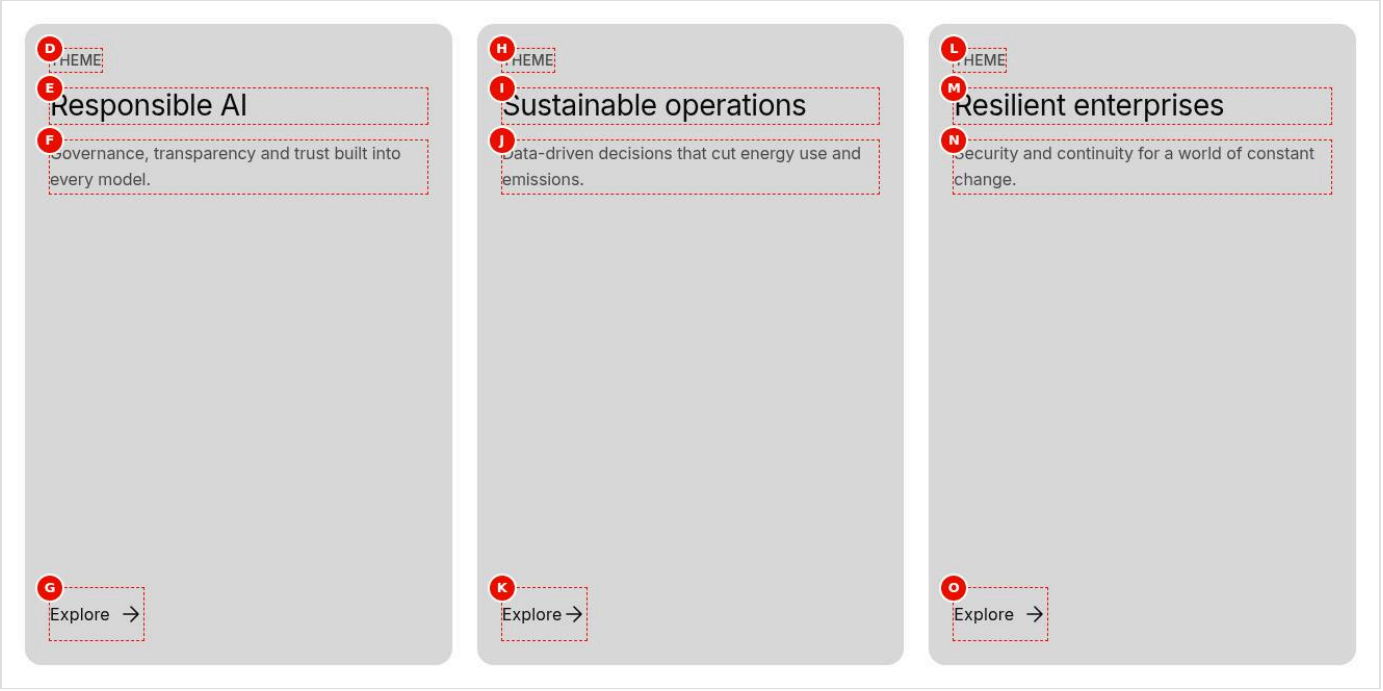


Figure: Light grey card colour — Three 2:3 panels in a 3-column row at 1440 px on light grey (#d9d9d9) with black text instead of the dark panel default: the kicker "Theme", the titles "Responsible AI", "Sustainable operations" and "Resilient enterprises", the descriptions and a black "Explore" link. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-colour-grey.jpg) (handover screenshot; rendering at the time it was taken).

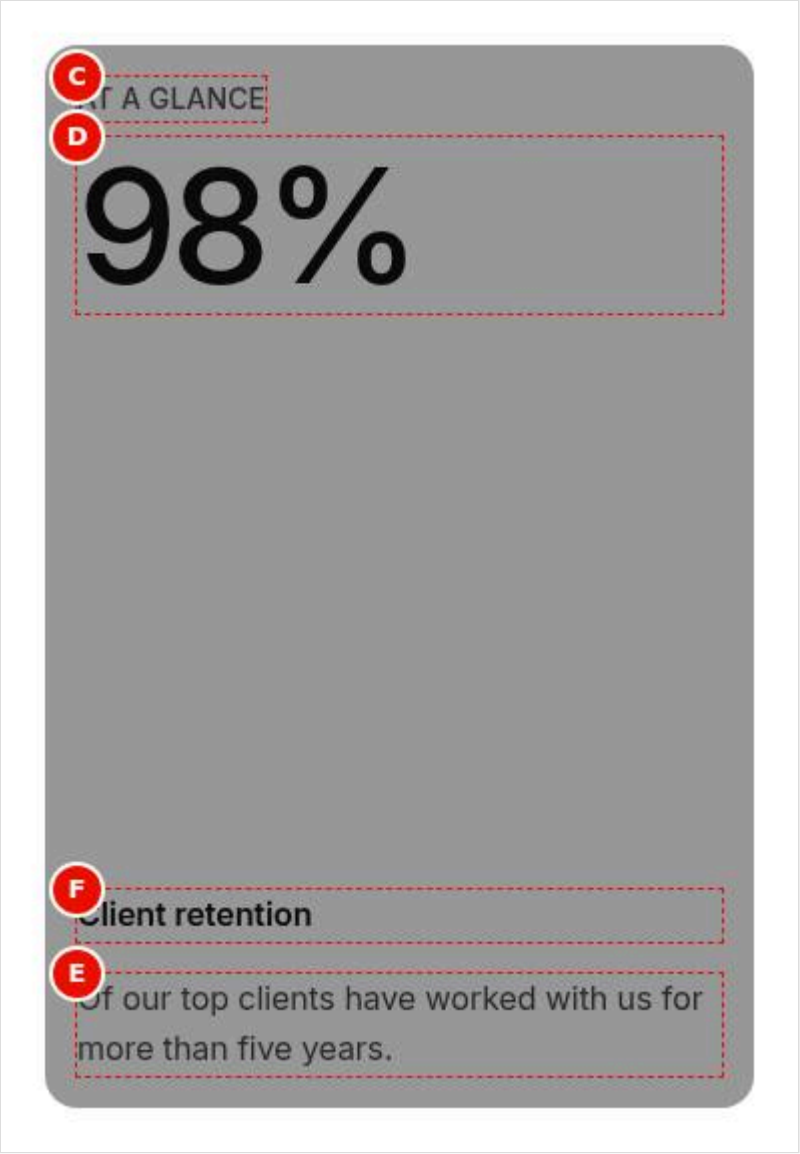


Figure: Mid grey card colour — At 402 px a single mid grey (#999) statistic card filling the column: "At a glance" at the top, the black number "98%", and at the bottom the label "Client retention" with a dark grey description. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-colour-mid-grey.jpg) (handover screenshot; rendering at the time it was taken).



Figure: Grey gradient card colour — At 402 px a single statistic card on the diagonal light blue-grey to warm grey gradient: "At a glance" at the top, the dark number "4bn", and at the bottom the label "Revenue" with its description. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-colour-gradient.jpg) (handover screenshot; rendering at the time it was taken).

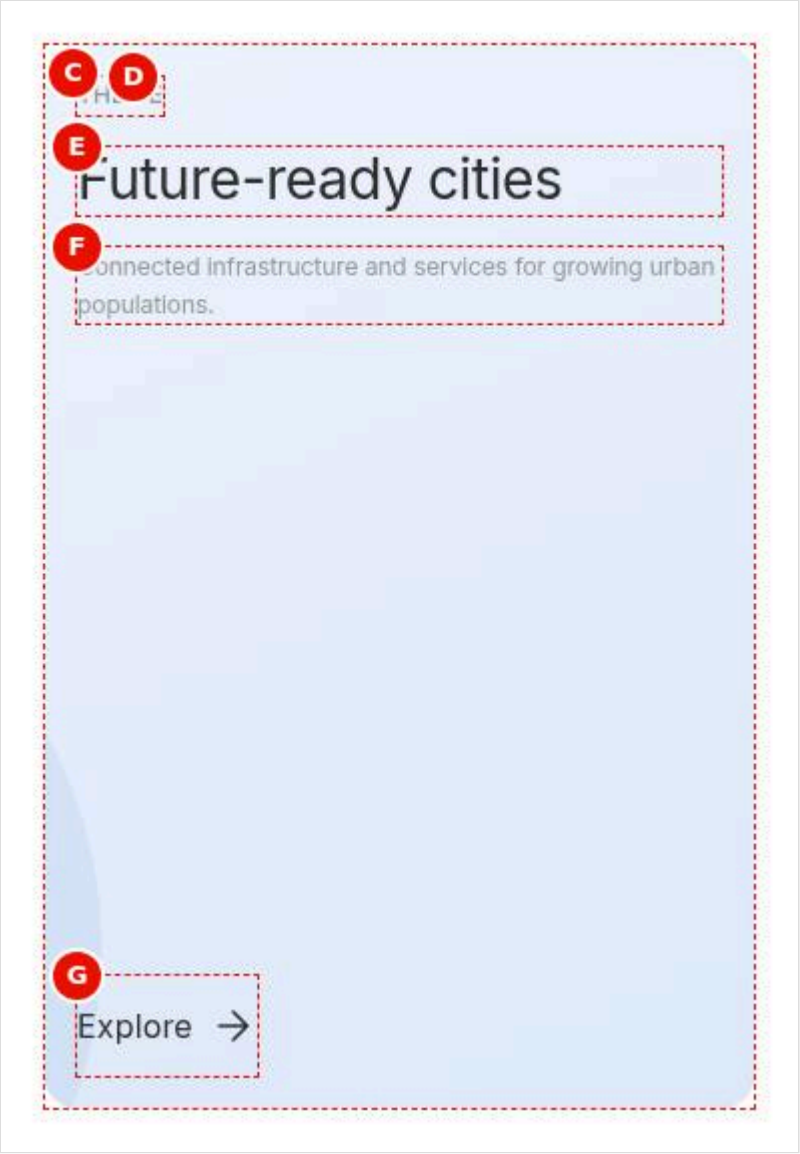


Figure: Image background card colour — At 402 px a single tall 2:3 panel whose image light.svg covers the whole card, with the kicker "Theme", the title "Future-ready cities", the description and the "Explore" link in dark text on top of the image (no overlay). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-colour-image.jpg) (handover screenshot; rendering at the time it was taken).

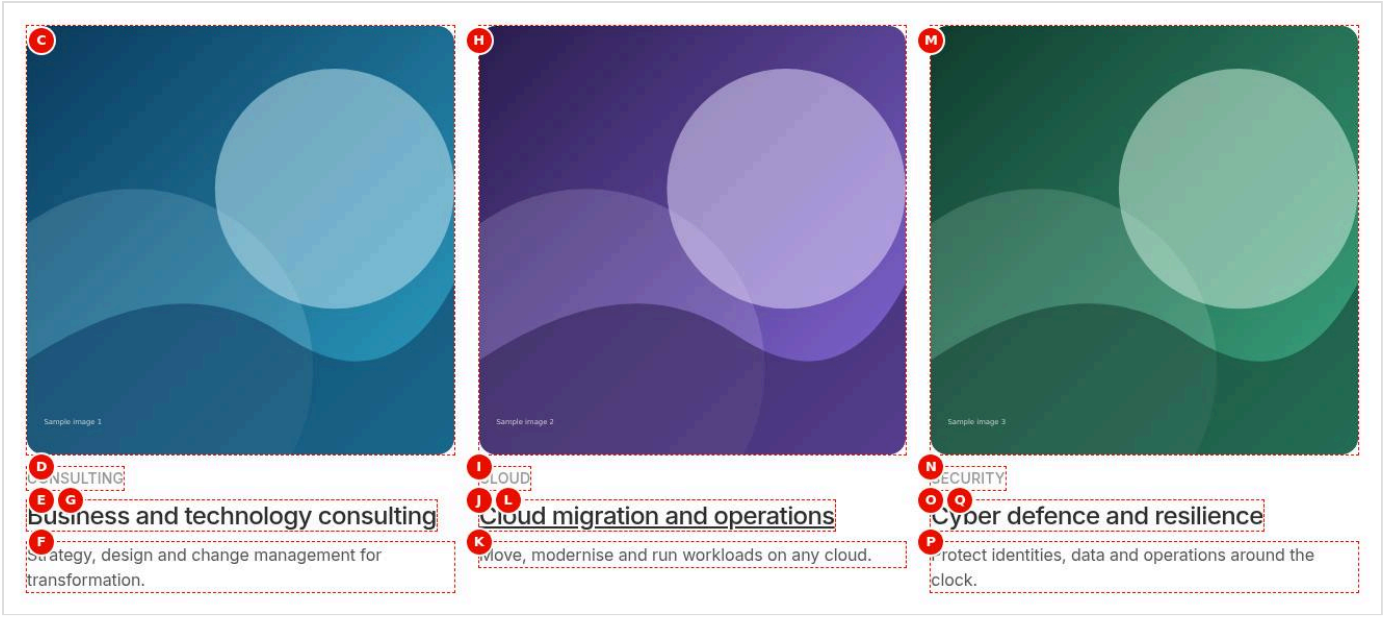


Figure: Square image ratio — Three image cards in a 3-column row at 1440 px with square images (square-1.svg to square-3.svg), the kickers "Consulting", "Cloud" and "Security", the titles "Business and technology consulting", "Cloud migration and operations" and "Cyber defence and resilience" and their descriptions; no meta row (no content type or date). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-ratio-1-1.jpg) (handover screenshot; rendering at the time it was taken).

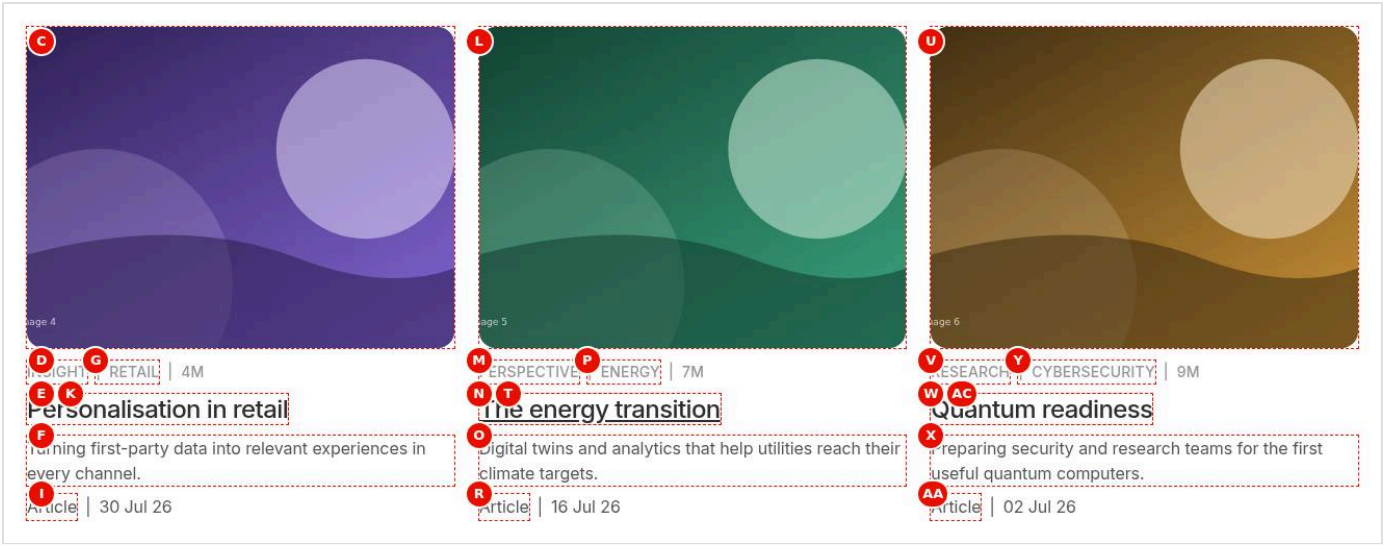


Figure: 4:3 image ratio — Three image cards in a 3-column row at 1440 px with 4:3 images (photo-4.svg to photo-6.svg), kickers with eyebrow, tag label and reading time, the titles "Personalisation in retail", "The energy transition" and "Quantum readiness", descriptions and the meta rows "Article" with "30 Jul 26", "16 Jul 26" and "02 Jul 26". (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-ratio-4-3.jpg) (handover screenshot; rendering at the time it was taken).

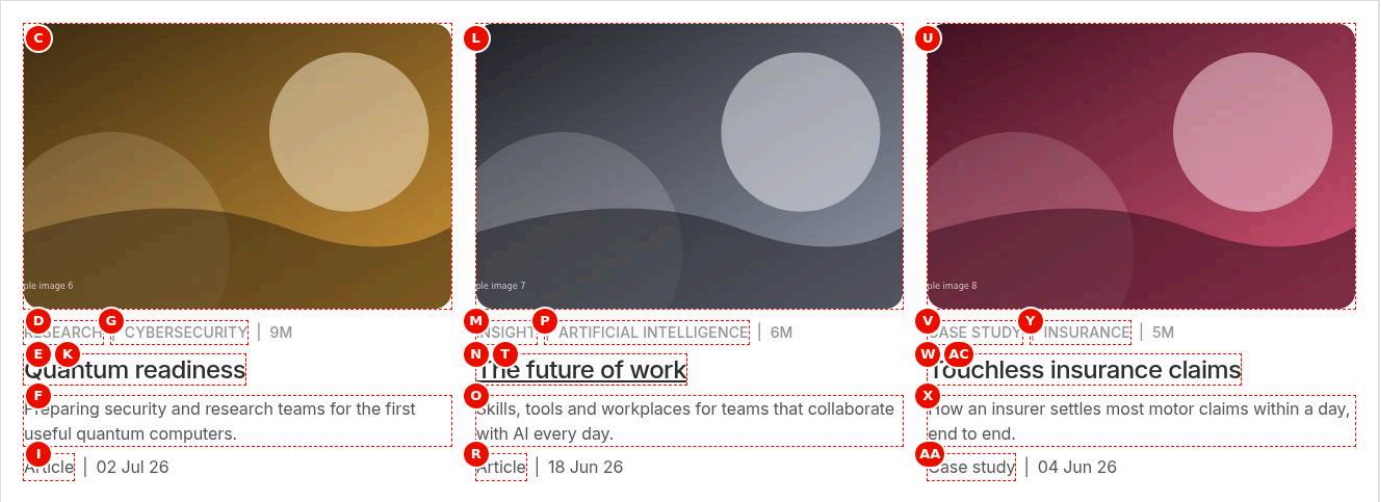


Figure: 3:2 image ratio — Three image cards in a 3-column row at 1440 px with 3:2 images (photo-6.svg to photo-8.svg), the titles "Quantum readiness", "The future of work" and "Touchless insurance claims", descriptions and the meta rows "Article" with "02 Jul 26", "Article" with "18 Jun 26" and "Case study" with "04 Jun 26". (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-ratio-3-2.jpg) (handover screenshot; rendering at the time it was taken).

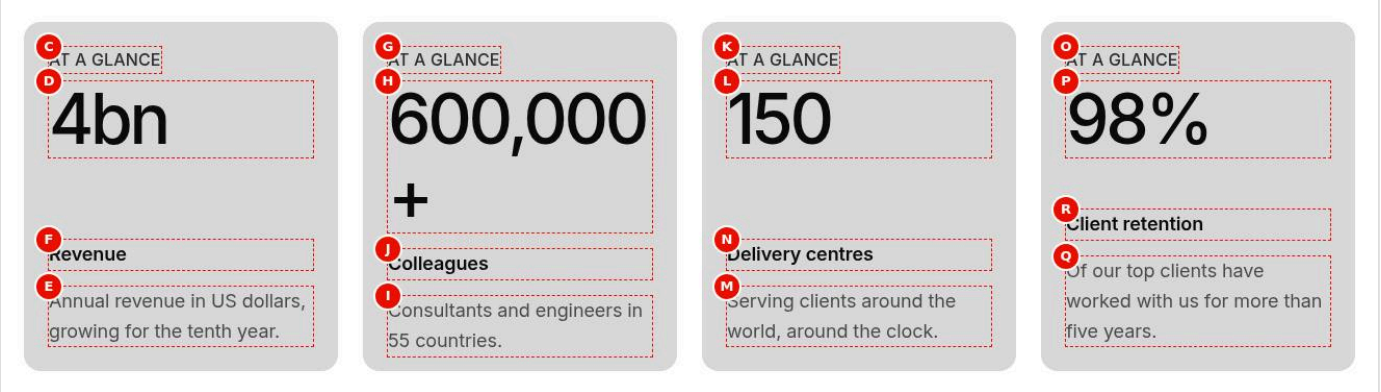


Figure: Compact panel height — Four nearly square (9:10) light grey statistic cards in a row at 1440 px instead of tall 2:3 ones, each with "At a glance", the number ("4bn", "600,000+", "150", "98%"), the label and the description. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-panel-compact.jpg) (handover screenshot; rendering at the time it was taken).

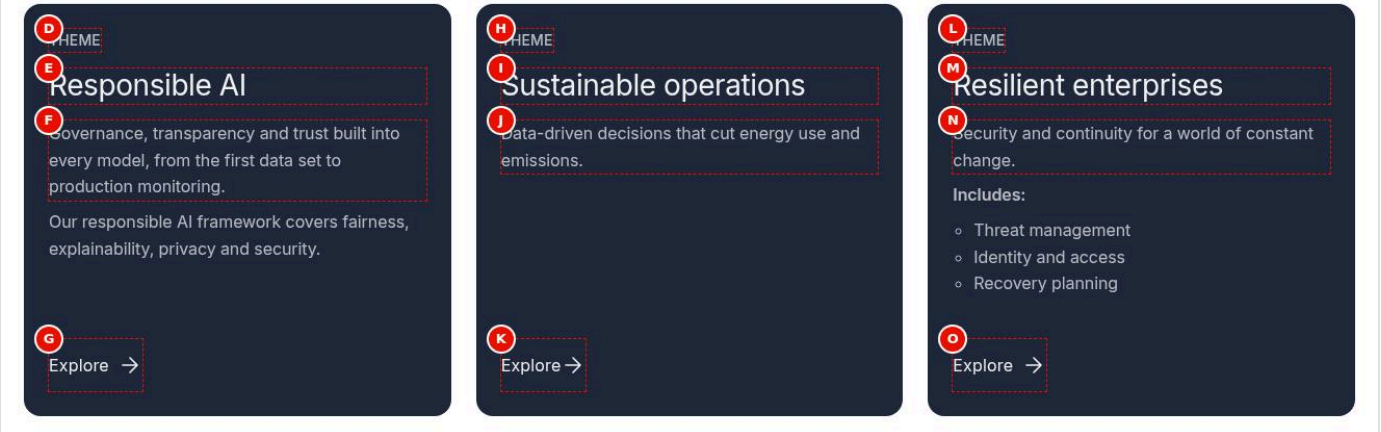


Figure: Content panel height — Three dark #20293c panels in a 3-column row at 1440 px whose height fits their text instead of the 2:3 shape; with Stretch cards to row height on they share the tallest height, each with the kicker "Theme", the title, the description and "Explore" at the bottom. (Sample content on the

preview; callout letters as in the Field Entry Table; file screenshots/cards-panel-content.jpg) (handover screenshot; rendering at the time it was taken).

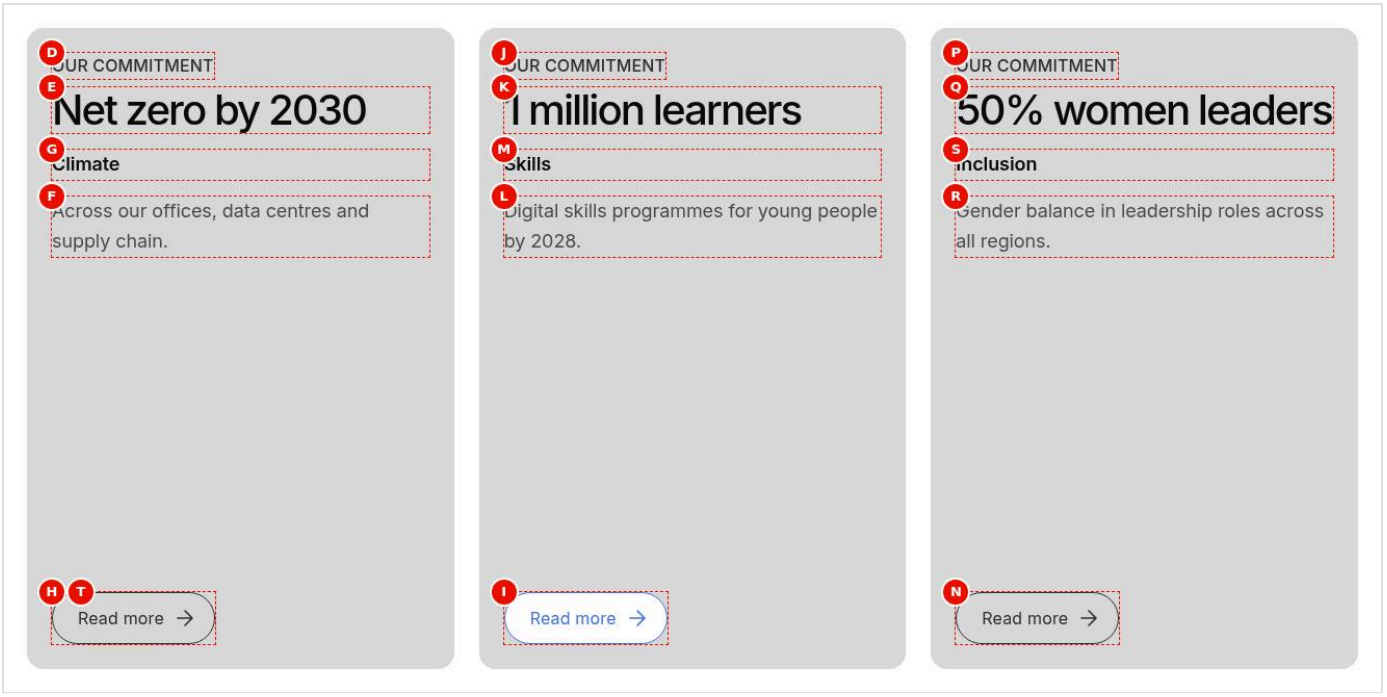


Figure: Compact statistic title — Three light grey statistic cards in a 3-column row at 1440 px with medium-size titles ("Net zero by 2030", "1 million learners", "50% women leaders") instead of huge numbers, the kicker "Our commitment", the labels "Climate", "Skills" and "Inclusion", descriptions and an outlined "Read more" button each. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-stat-compact.jpg) (handover screenshot; rendering at the time it was taken).

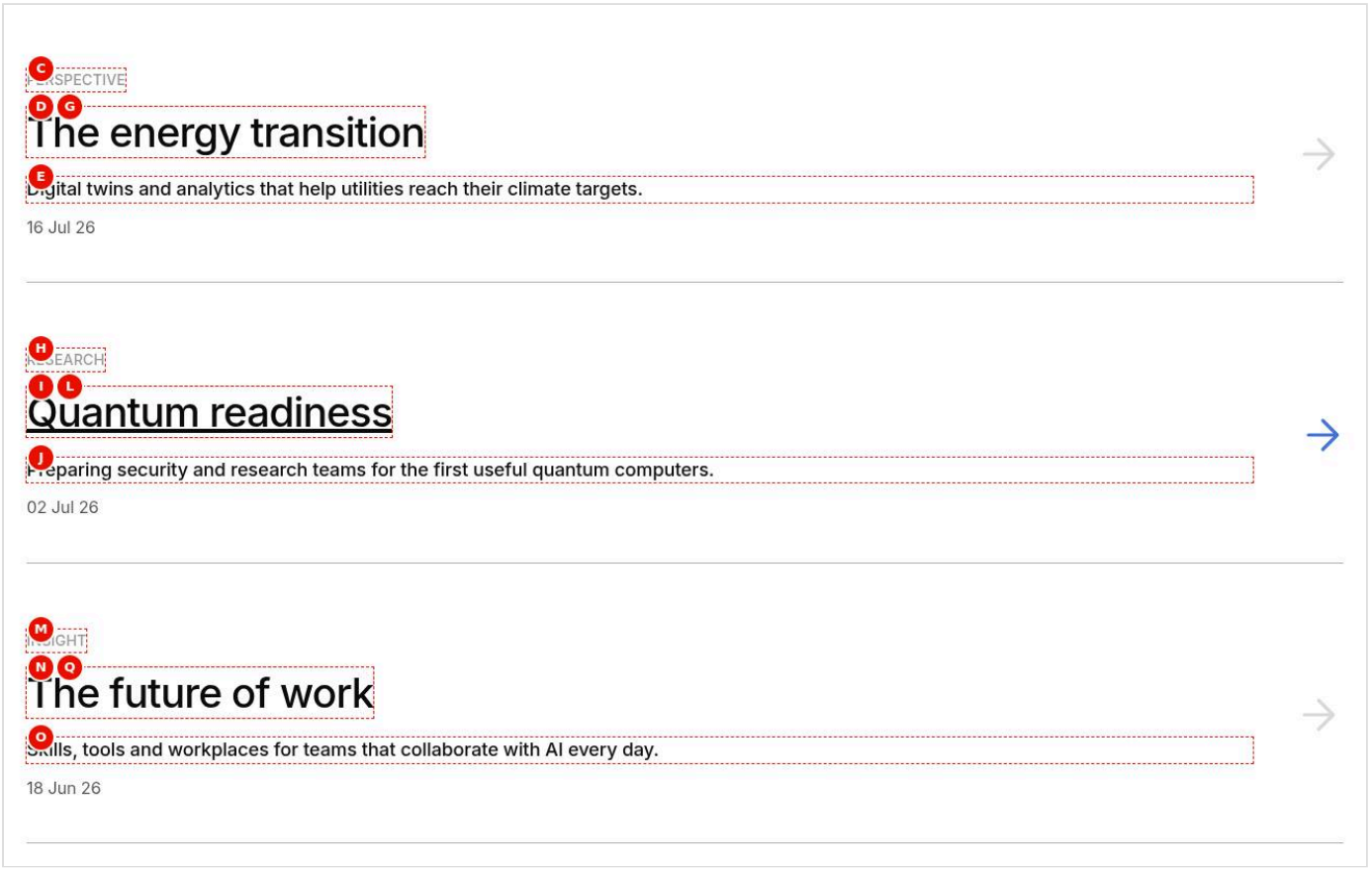


Figure: Dividers between rows — Three rows of links at 1440 px separated by thin grey (#b2b2b2) lines, each with the eyebrow ("Perspective", "Research", "Insight"), the large title ("The energy transition", "Quantum readiness", "The future of work"), the description, the date and a grey arrow on the right. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-dividers.jpg) (handover screenshot; rendering at the time it was taken).

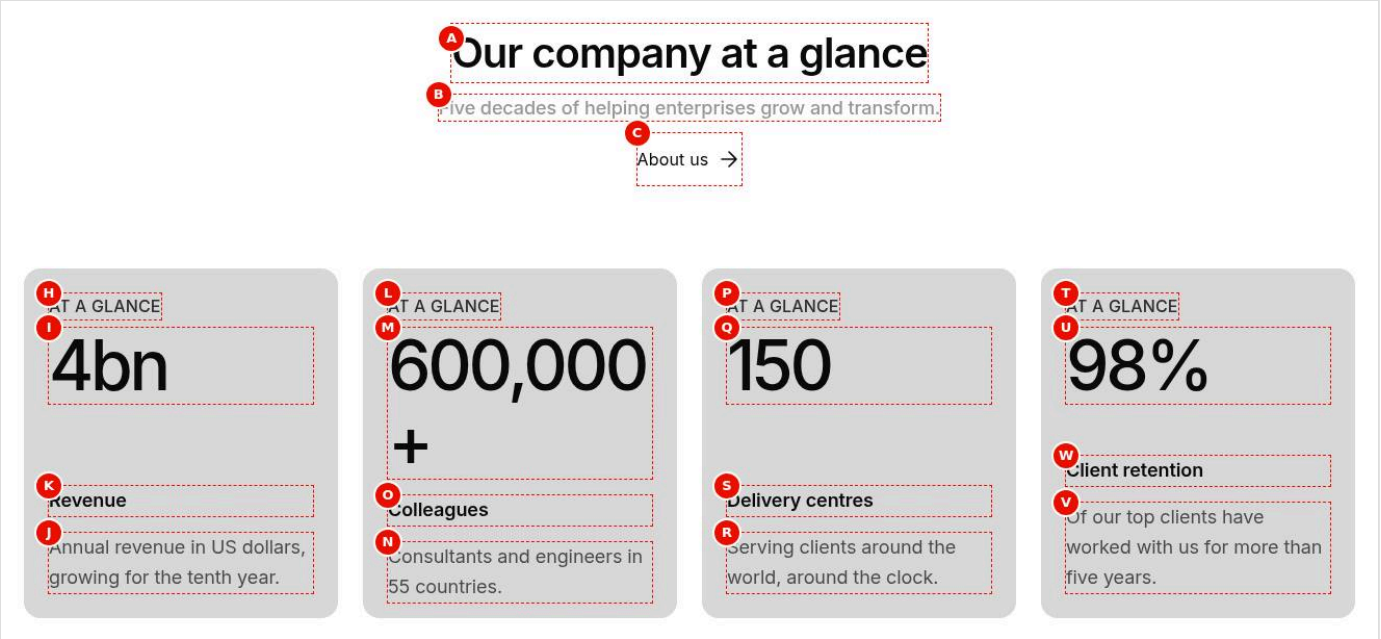


Figure: Centred header — At 1440 px the title "Our company at a glance", its description and the "About us" text link centred above a row of four compact (9:10) light grey statistic cards ("4bn", "600,000+", "150", "98%"). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-header-centred.jpg) (handover screenshot; rendering at the time it was taken).

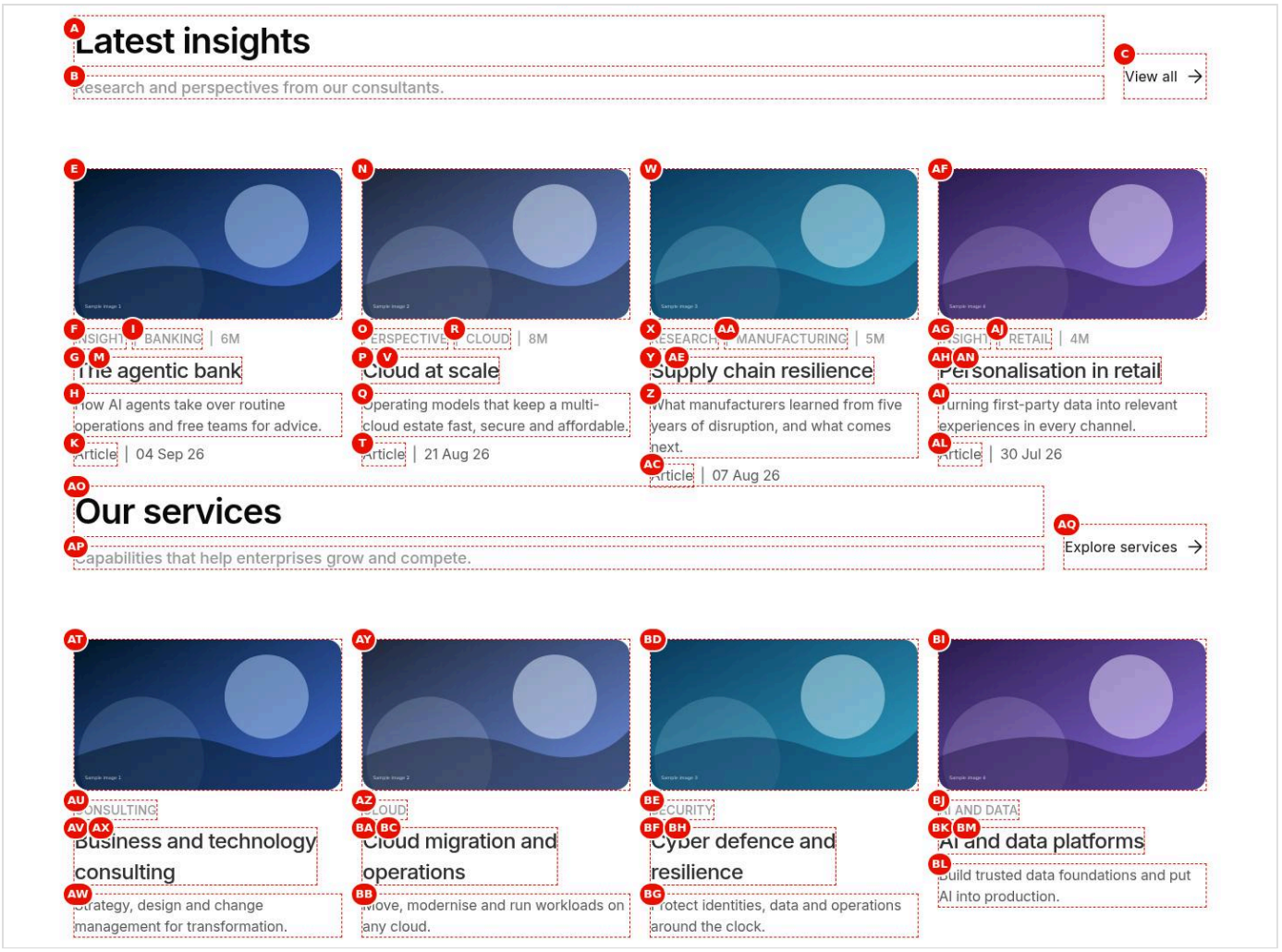


Figure: Header call to action — One section at 1440 px with two Cards blocks in a single capture: "Latest insights" with a "View all" text link on the right above four insight image cards, then "Our services" with a filled primary "Explore services" button above four service image cards. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-header-cta.jpg) (handover screenshot; rendering at the time it was taken).

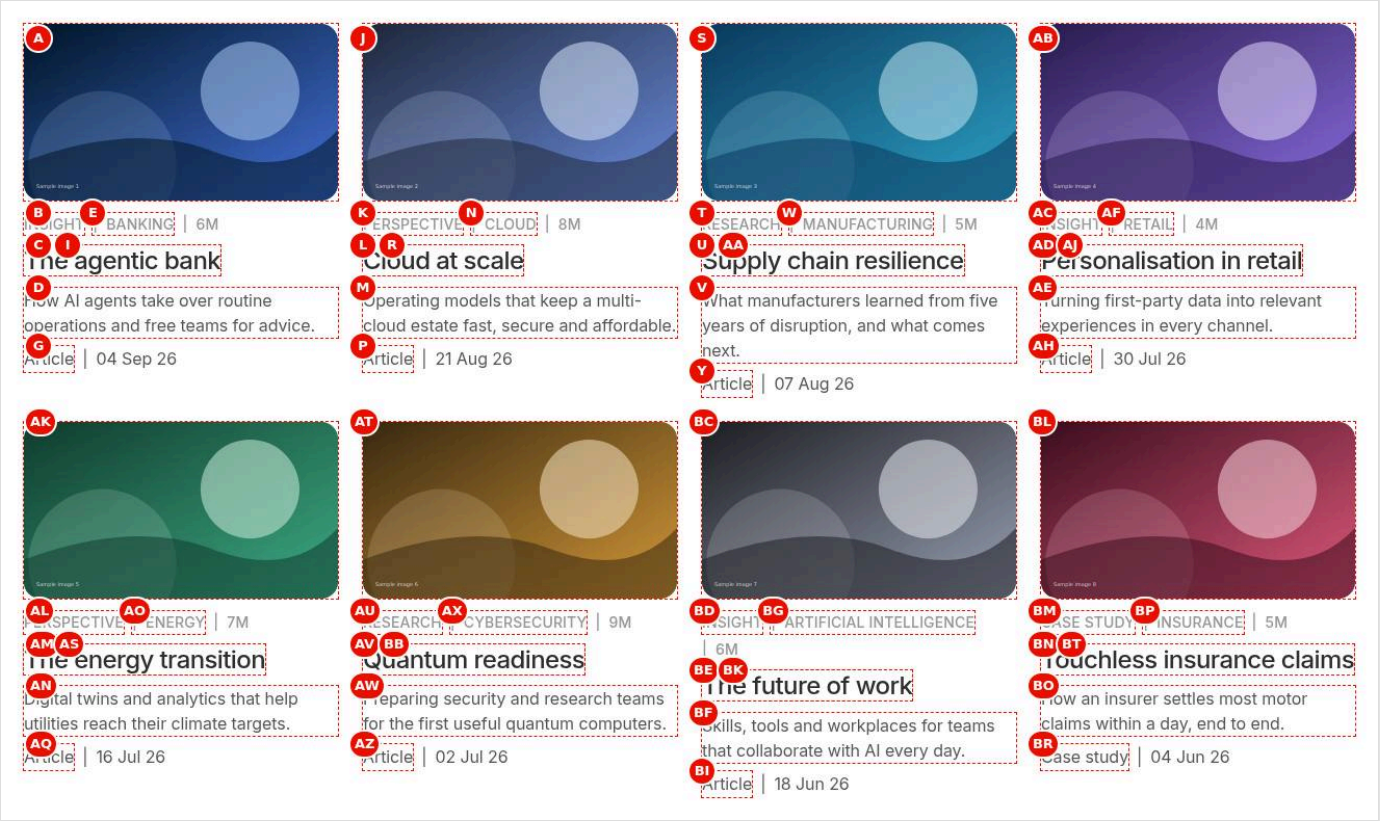


Figure: Grid on every breakpoint — At 1440 px eight image cards in two rows of four (default Grid, 4 columns), from "The agentic bank" to "Touchless insurance claims"; only the desktop layout is captured (below 600 px the cards would stack in one column). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-grid-all.jpg) (handover screenshot; rendering at the time it was taken).

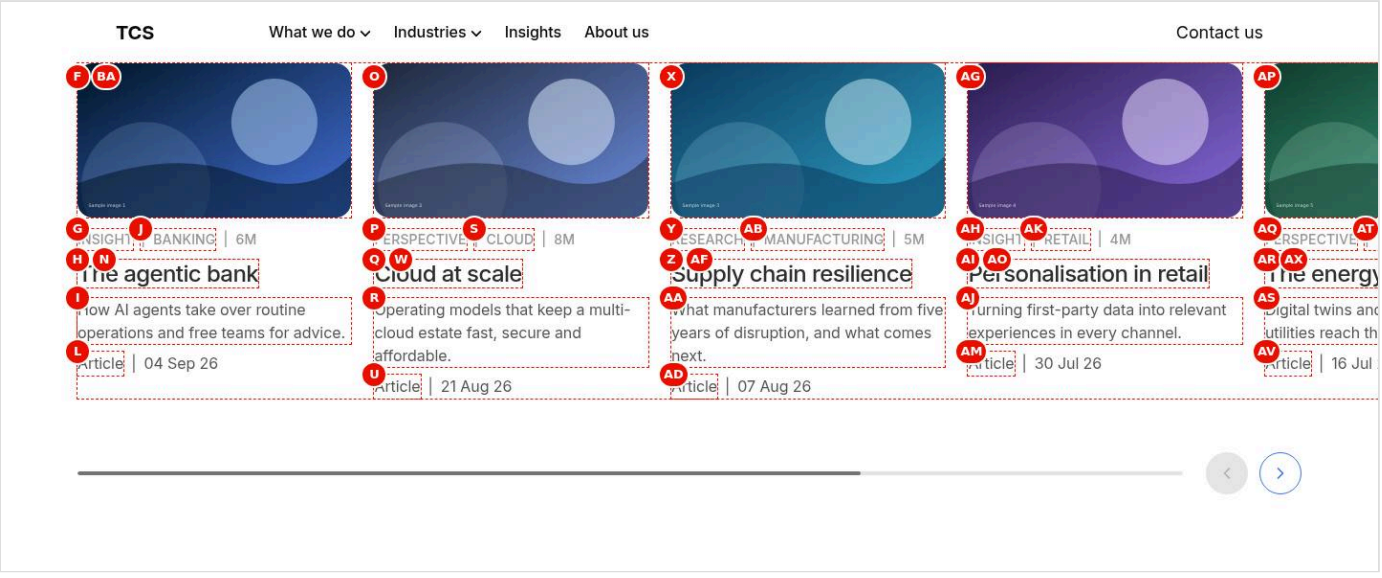


Figure: Horizontal scroll on every breakpoint — At 1440 px a scrolling row with 4.2 image cards visible, the fifth cut off at the right screen edge (bleed), with the scroll indicator and the round previous / next buttons below (Previous disabled at the start); only the desktop layout is captured. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-scroll-all.jpg) (handover screenshot; rendering at the time it was taken).

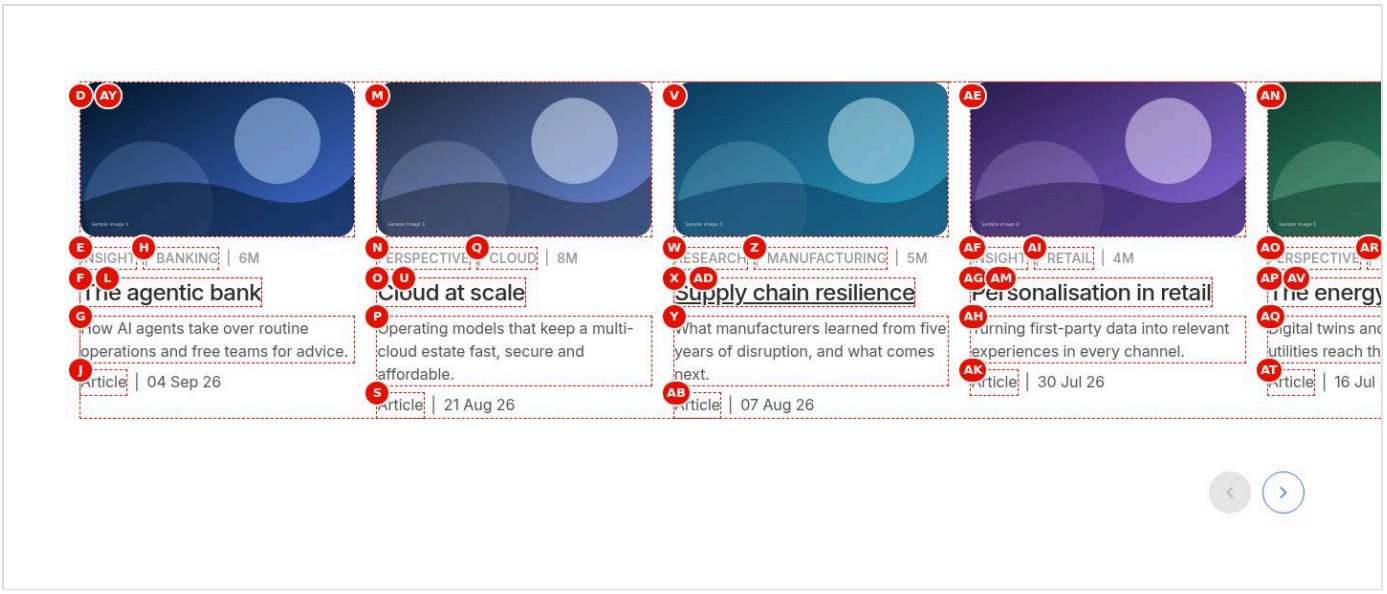


Figure: Desktop-only scroll — At 1440 px a scrolling row of the six image cards with 4.2 visible, the fifth cut off at the screen edge, and the previous / next buttons at the bottom right; only the desktop layout is captured (tablet would be a 2-column grid, mobile one column). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-scroll-desktop-only.jpg) (handover screenshot; rendering at the time it was taken).

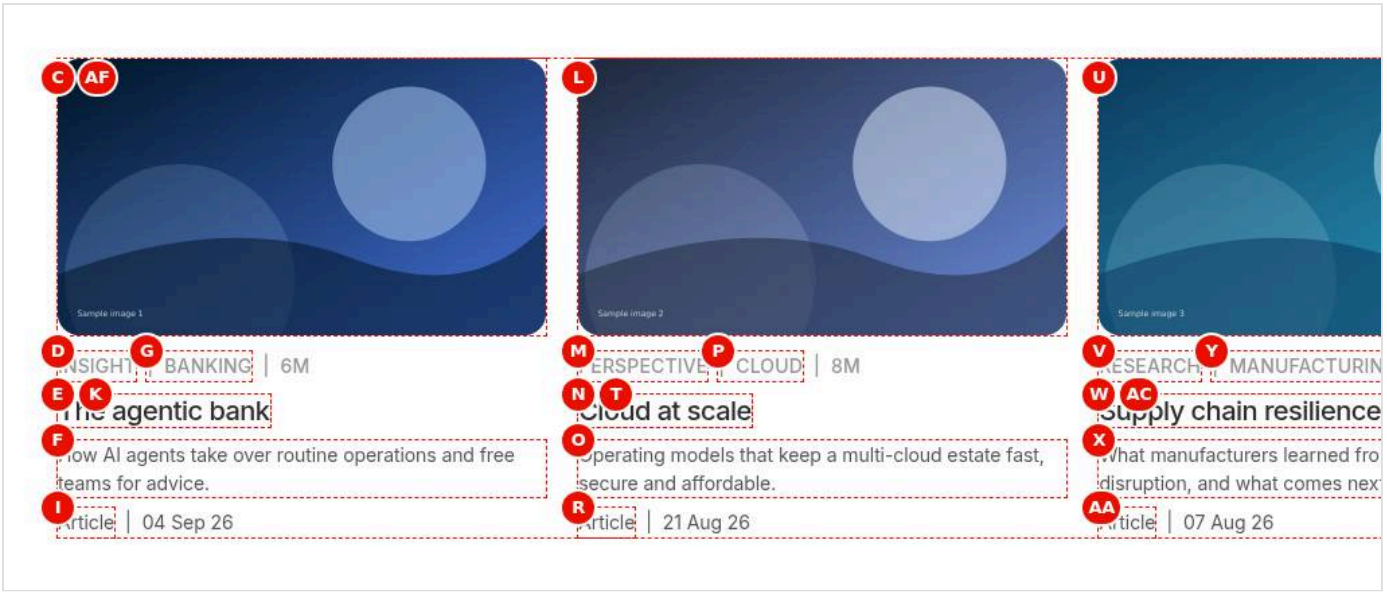


Figure: Tablet-only scroll — At 1024 px a scrolling row with 2.5 image cards visible ("The agentic bank", "Cloud at scale" and half of "Supply chain resilience"), cut off at the right edge, without buttons or indicator (tablet defaults off). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-scroll-tablet-only.jpg) (handover screenshot; rendering at the time it was taken).

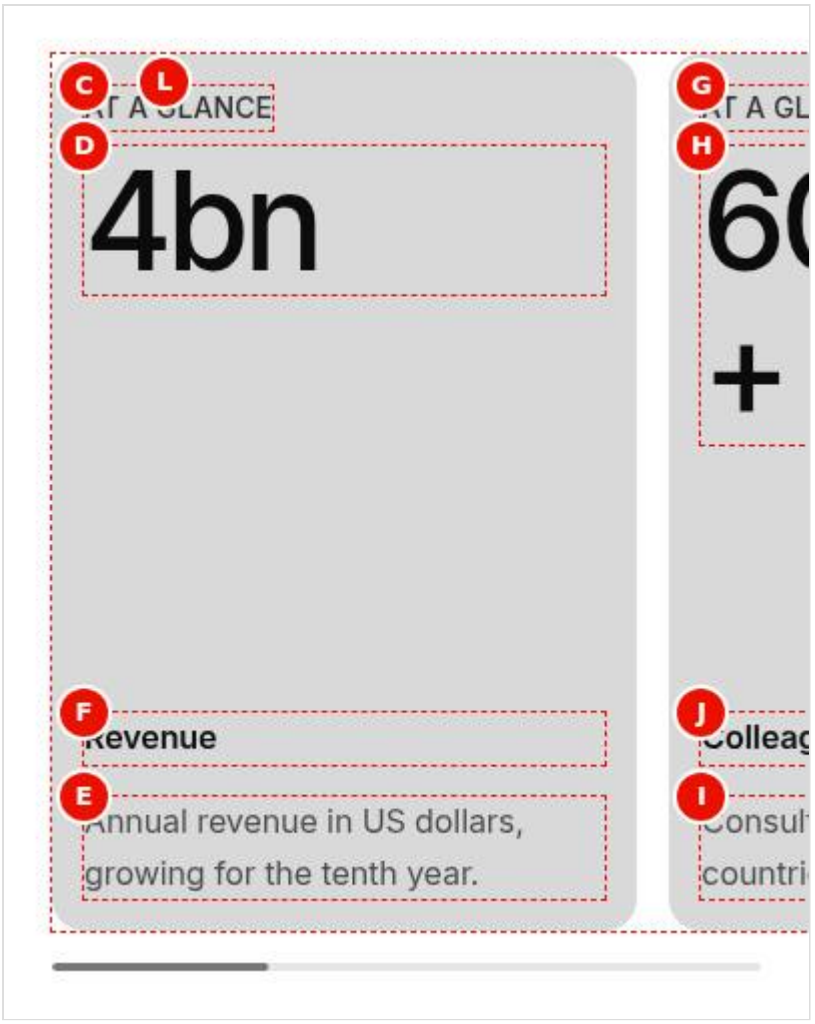


Figure: Mobile-only scroll — At 402 px a rail of light grey statistic cards with 1.2 visible: "4bn" and part of "600,000+" peeking at the right edge, with the grey scroll indicator below. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-scroll-mobile-only.jpg) (handover screenshot; rendering at the time it was taken).

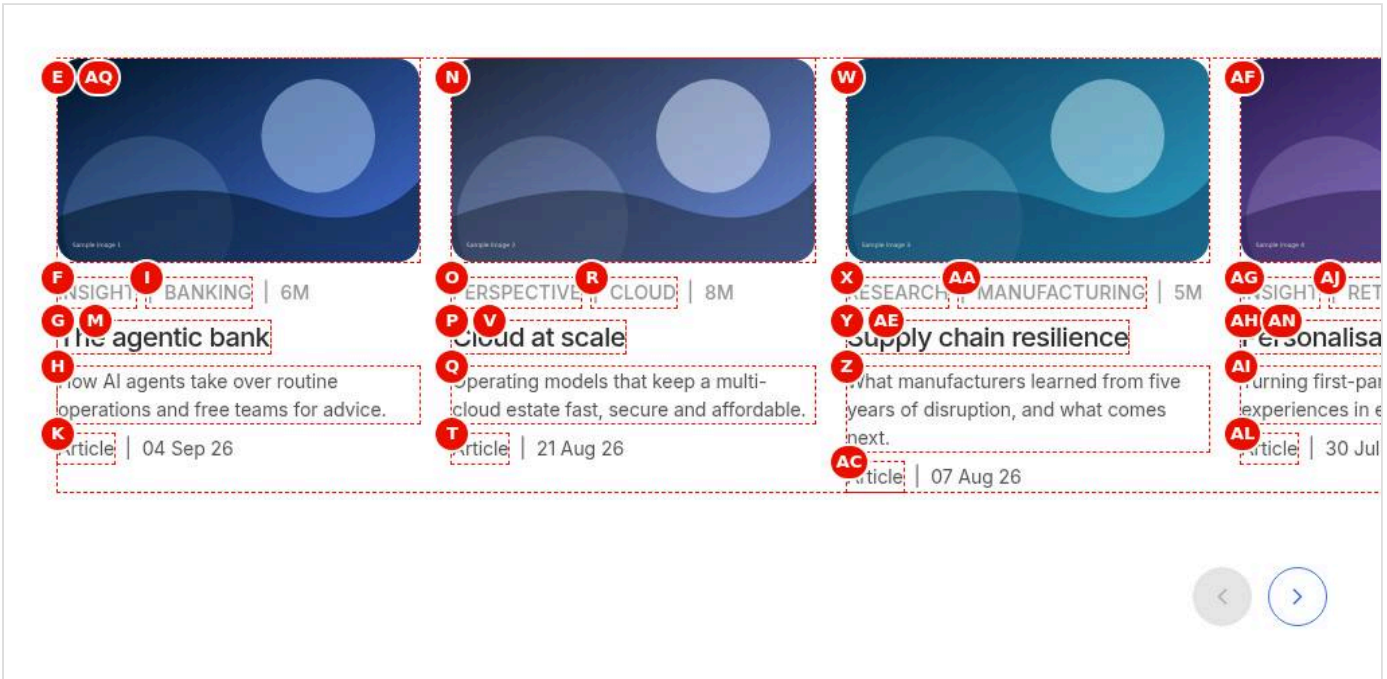


Figure: Desktop and tablet scroll — At 1024 px a scrolling row with 3.3 image cards visible, the fourth cut off at the right edge, and the previous / next buttons at the bottom right (Previous disabled at the start). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-scroll-desktop-tablet.jpg) (handover screenshot; rendering at the time it was taken).

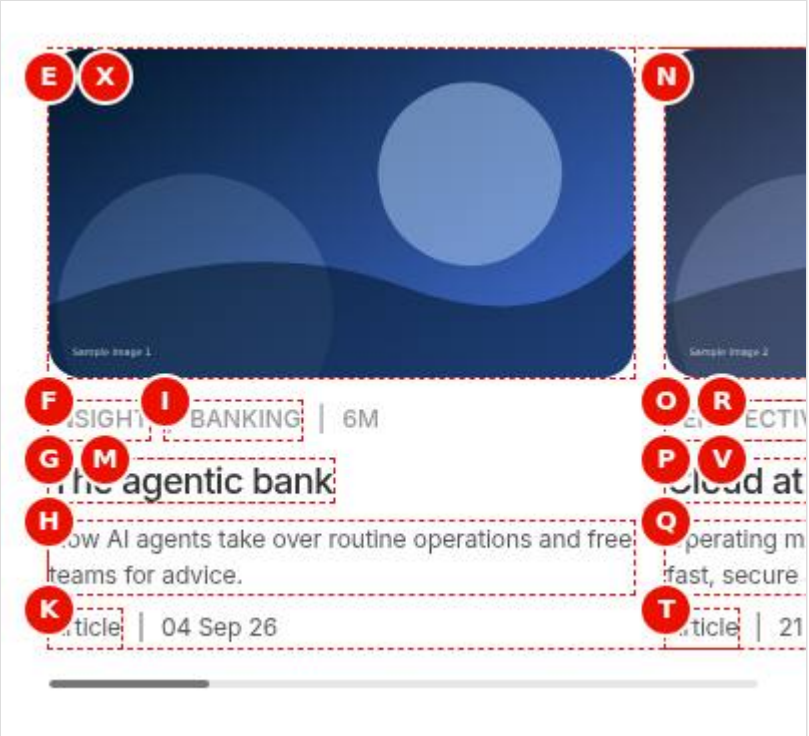


Figure: Desktop and mobile scroll — At 402 px a rail of image cards with 1.2 visible, "The agentic bank" and part of "Cloud at scale" at the right edge, with the grey scroll indicator below (mobile buttons off by default). (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-scroll-desktop-mobile.jpg) (handover screenshot; rendering at the time it was taken).

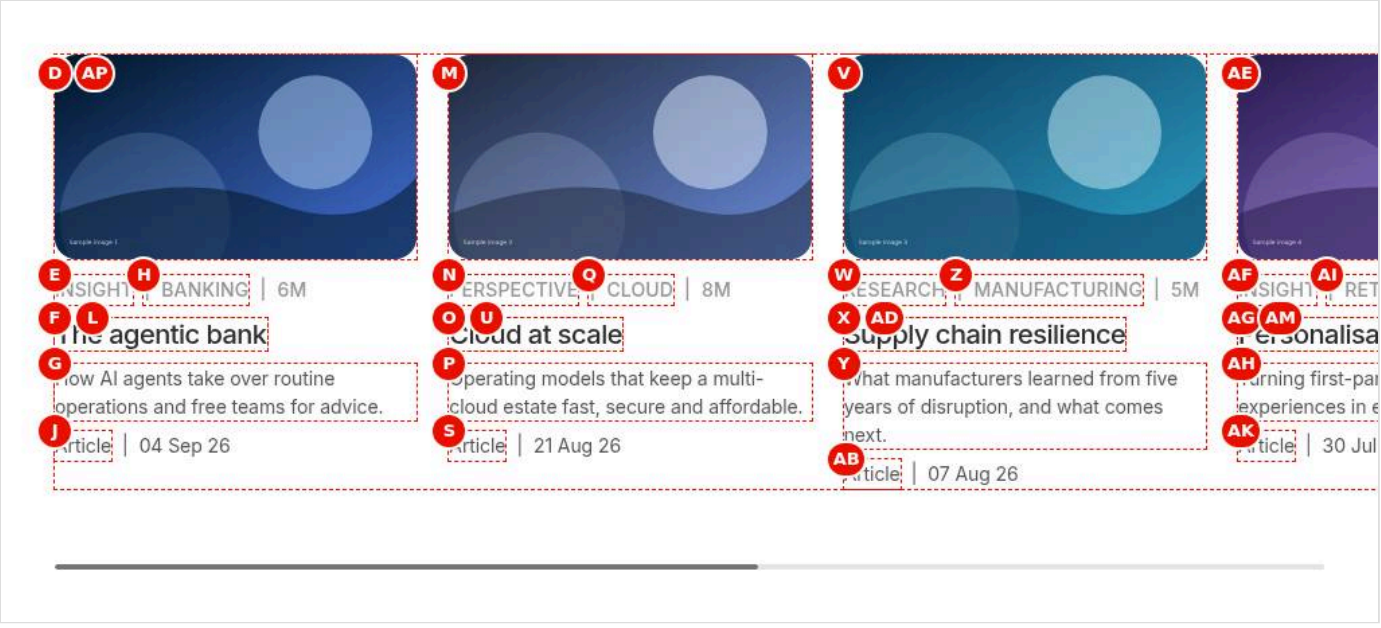


Figure: Tablet and mobile scroll — At 1024 px a scrolling row with 3.3 image cards visible, the fourth cut off at the right edge, and a thin grey scroll indicator below; no previous / next buttons. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-scroll-tablet-mobile.jpg) (handover screenshot; rendering at the time it was taken).

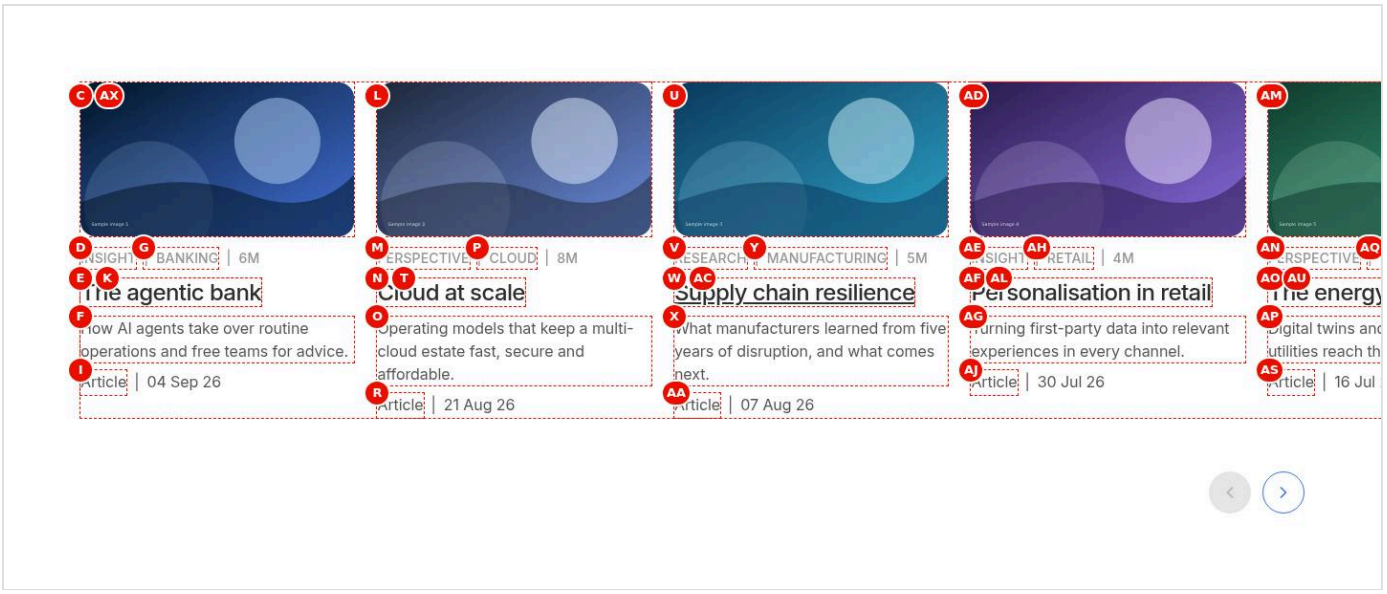


Figure: Previous and next buttons — At 1440 px a scrolling row of six image cards with 4.2 visible and two round chevron icon buttons at the bottom right under the row; the Previous button is disabled at the start and Next is enabled. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-previous-next.jpg) (handover screenshot; rendering at the time it was taken).

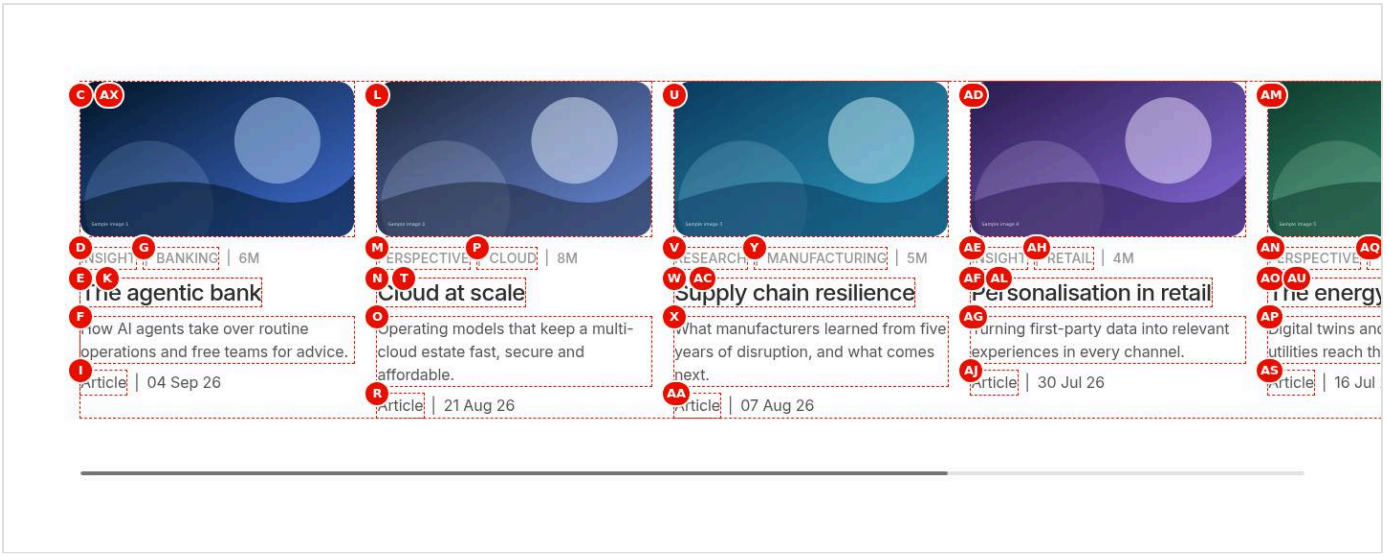


Figure: Scroll indicator — At 1440 px a scrolling row of six image cards with 4.2 visible and below it a 4-pixel light grey track with a darker grey thumb at the start whose width shows the visible share; no buttons. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-scroll-indicator.jpg) (handover screenshot; rendering at the time it was taken).

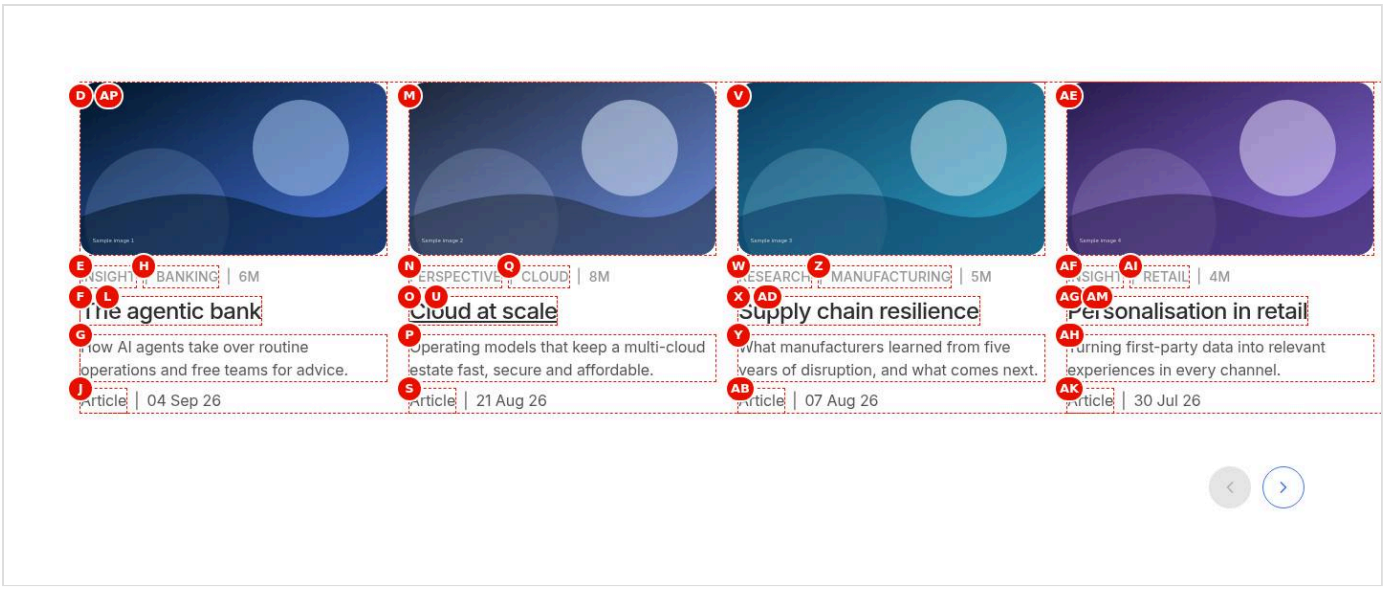


Figure: Fixed card width — At 1440 px a scrolling row of fixed 320 px (20rem) image cards, about four visible with the last cut off at the screen edge, and the previous / next buttons below. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-fixed-card-width.jpg) (handover screenshot; rendering at the time it was taken).

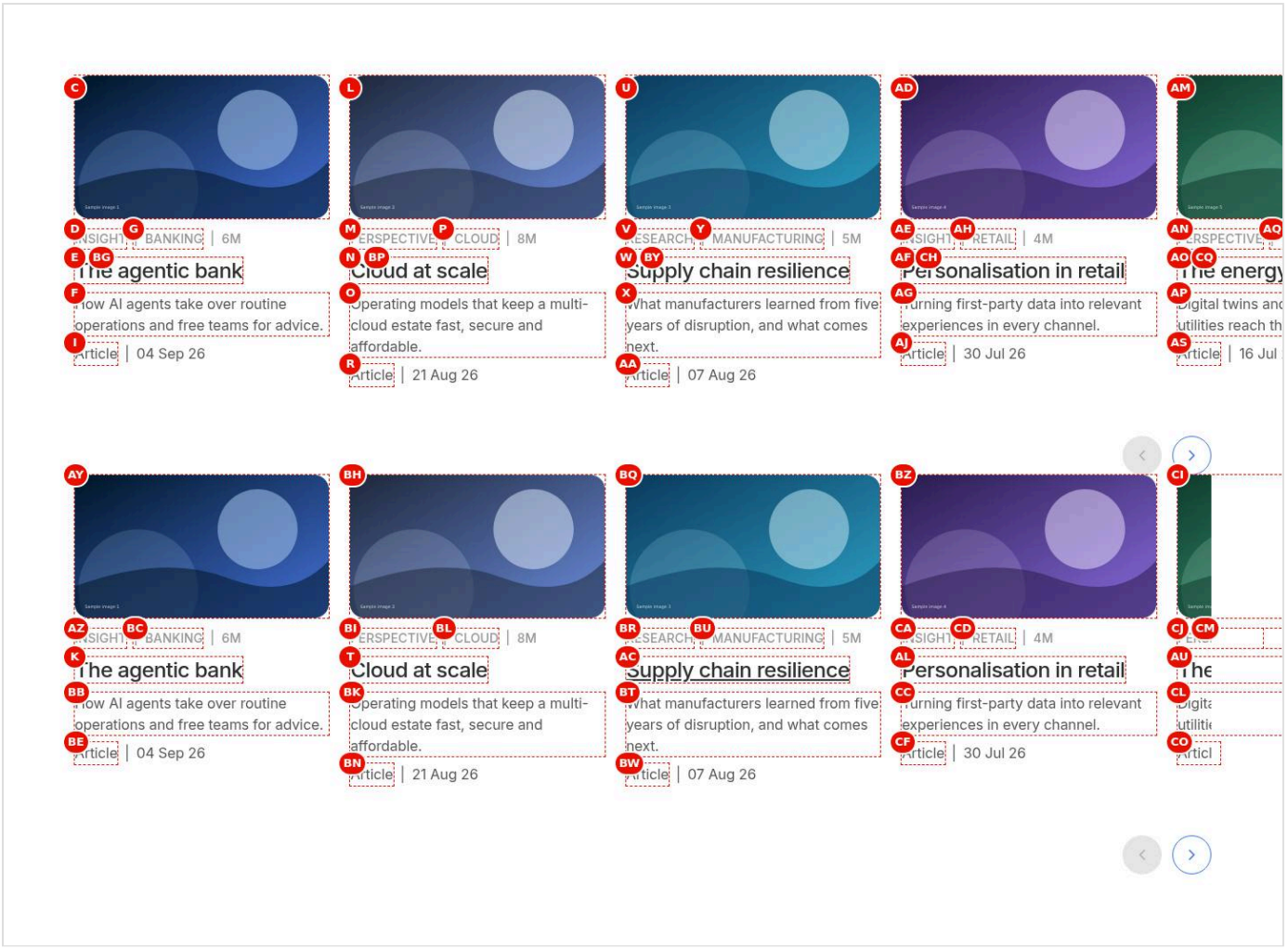


Figure: Scroller inside the content width — One section at 1440 px with two desktop rails of five image cards in a single capture: in the first (bleed) the last visible card is cut by the screen edge; in the second (Inside the content width) it is cut at the section's right padding. Both have previous / next buttons.

(Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-edge-none.jpg) (handover screenshot; rendering at the time it was taken).

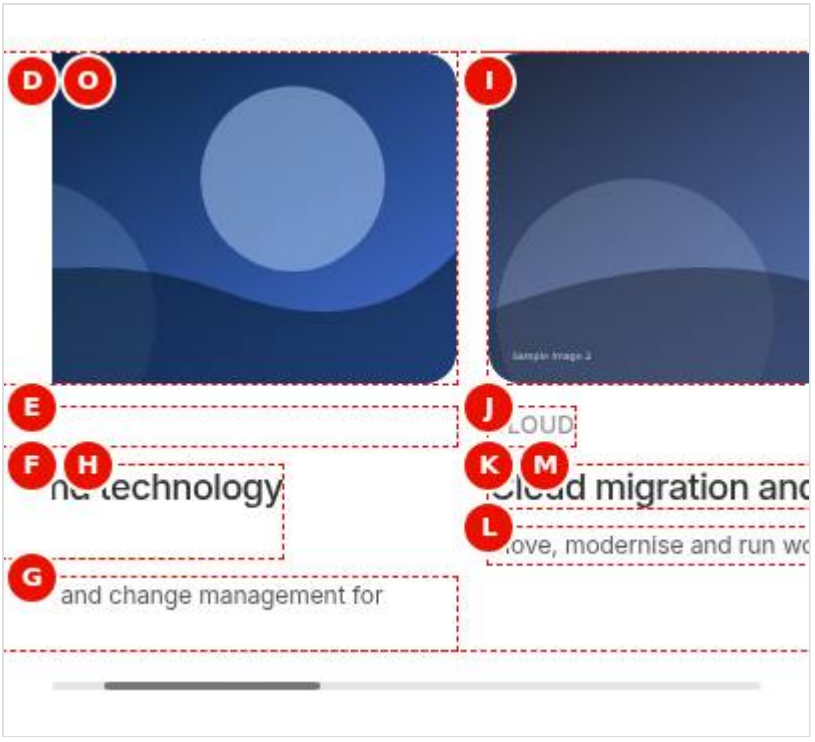


Figure: Centre snap alignment — At 402 px a rail of service image cards opened at the second card (Initial item index 1): "Cloud migration and operations" centred, with parts of the first and third cards visible on both sides, and the scroll indicator below. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-snap-center.jpg) (handover screenshot; rendering at the time it was taken).

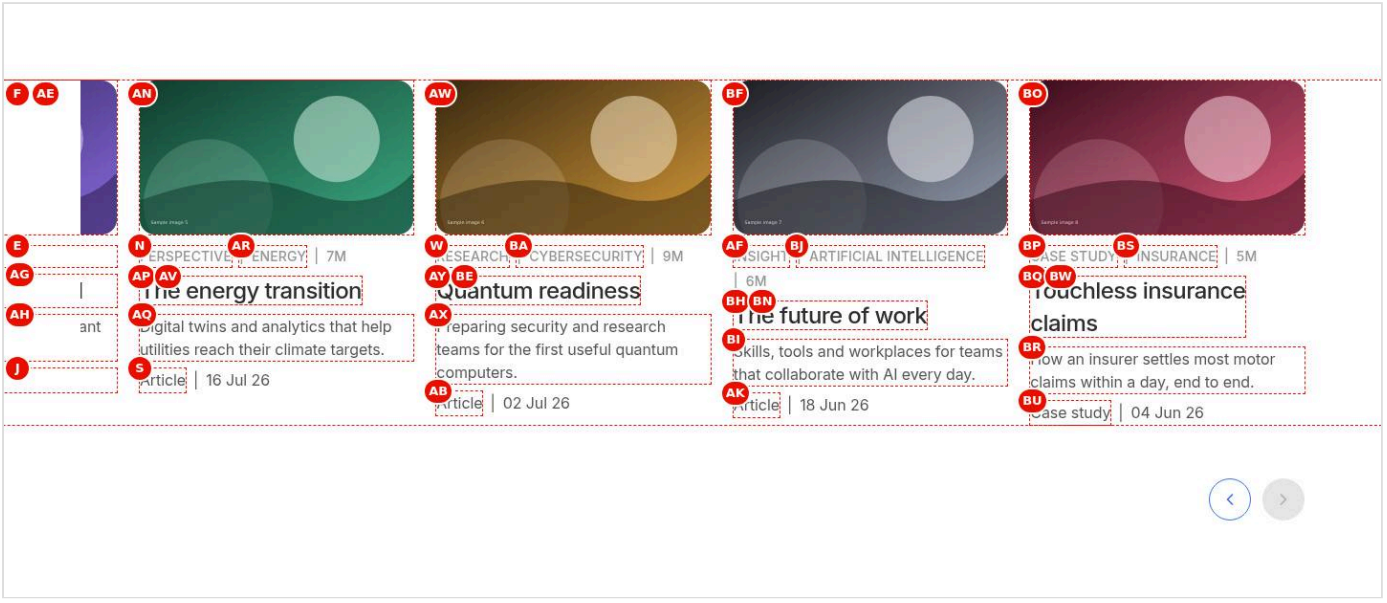


Figure: Viewport scroll step — At 1440 px the rail of eight image cards after one click on Next (a single capture, not before and after): the next set of cards fills the view and the Previous button is enabled. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-step-viewport.jpg) (handover screenshot; rendering at the time it was taken).

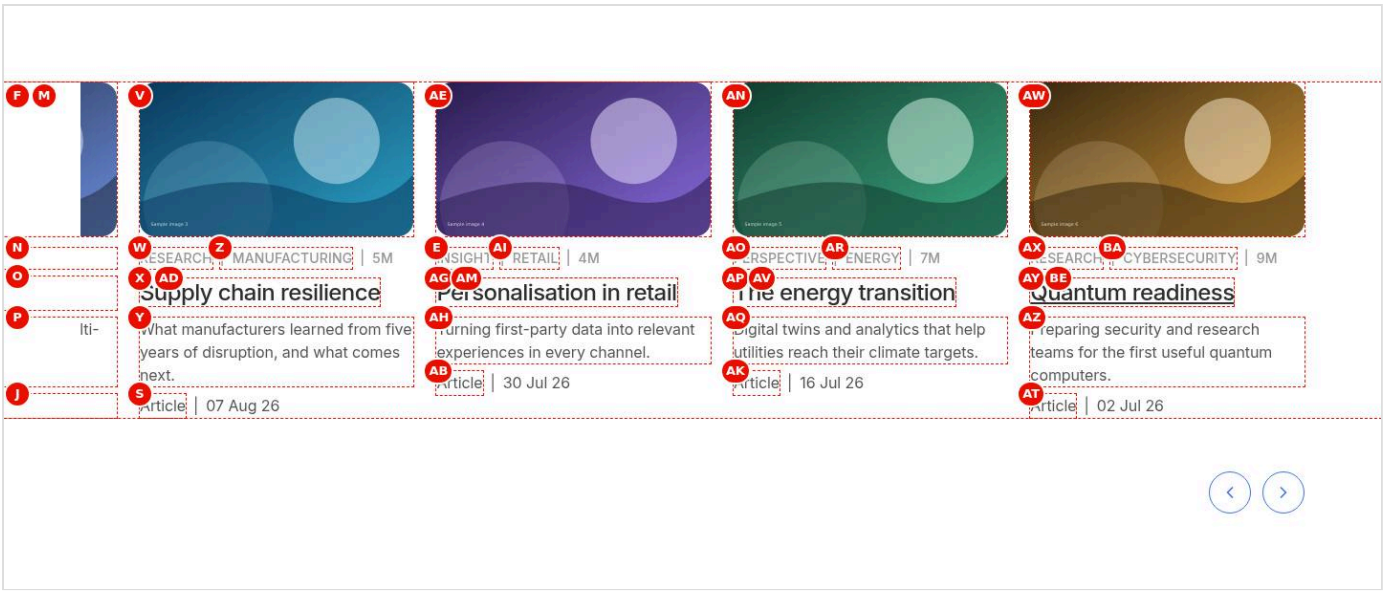


Figure: Loop — At 1440 px the rail of six image cards scrolled to its end ("Quantum readiness" visible at the right) with the Next button still enabled because Loop is on; the capture shows the end state, not the return to the first card. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-loop.jpg) (handover screenshot; rendering at the time it was taken).

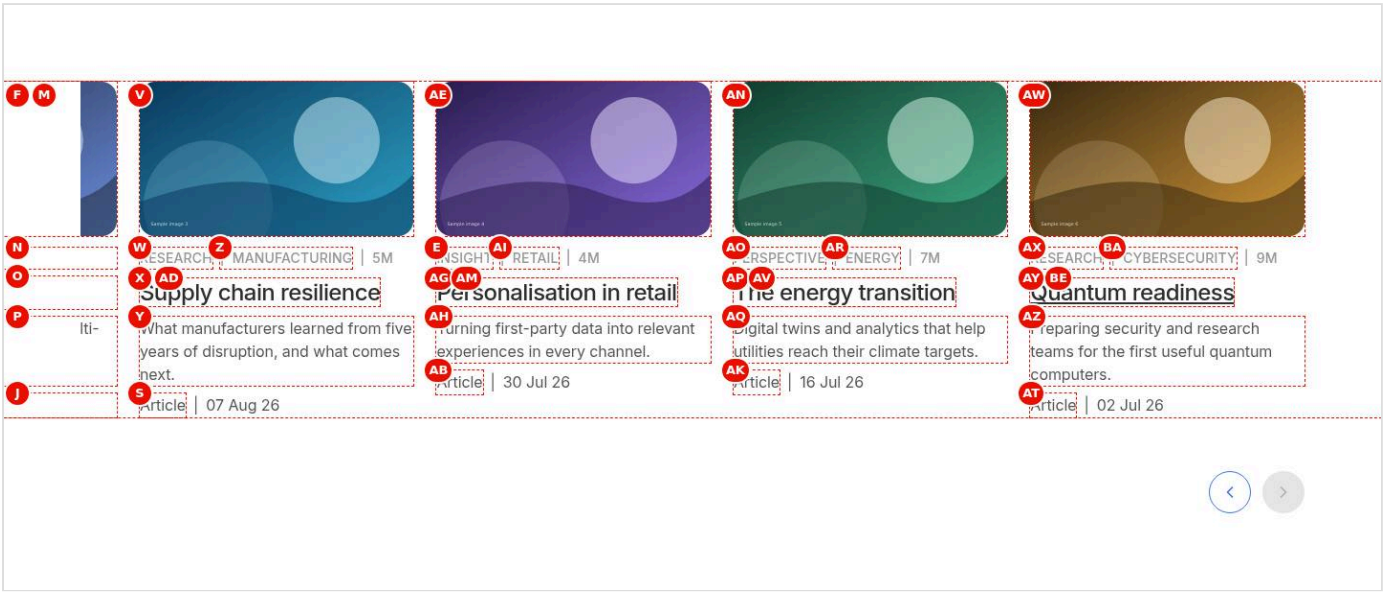


Figure: Initial item — At 1440 px a rail of six image cards that opens scrolled to its third card, "Supply chain resilience", at the left edge (Initial item index 2), with the Previous button enabled. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-initial-item.jpg) (handover screenshot; rendering at the time it was taken).

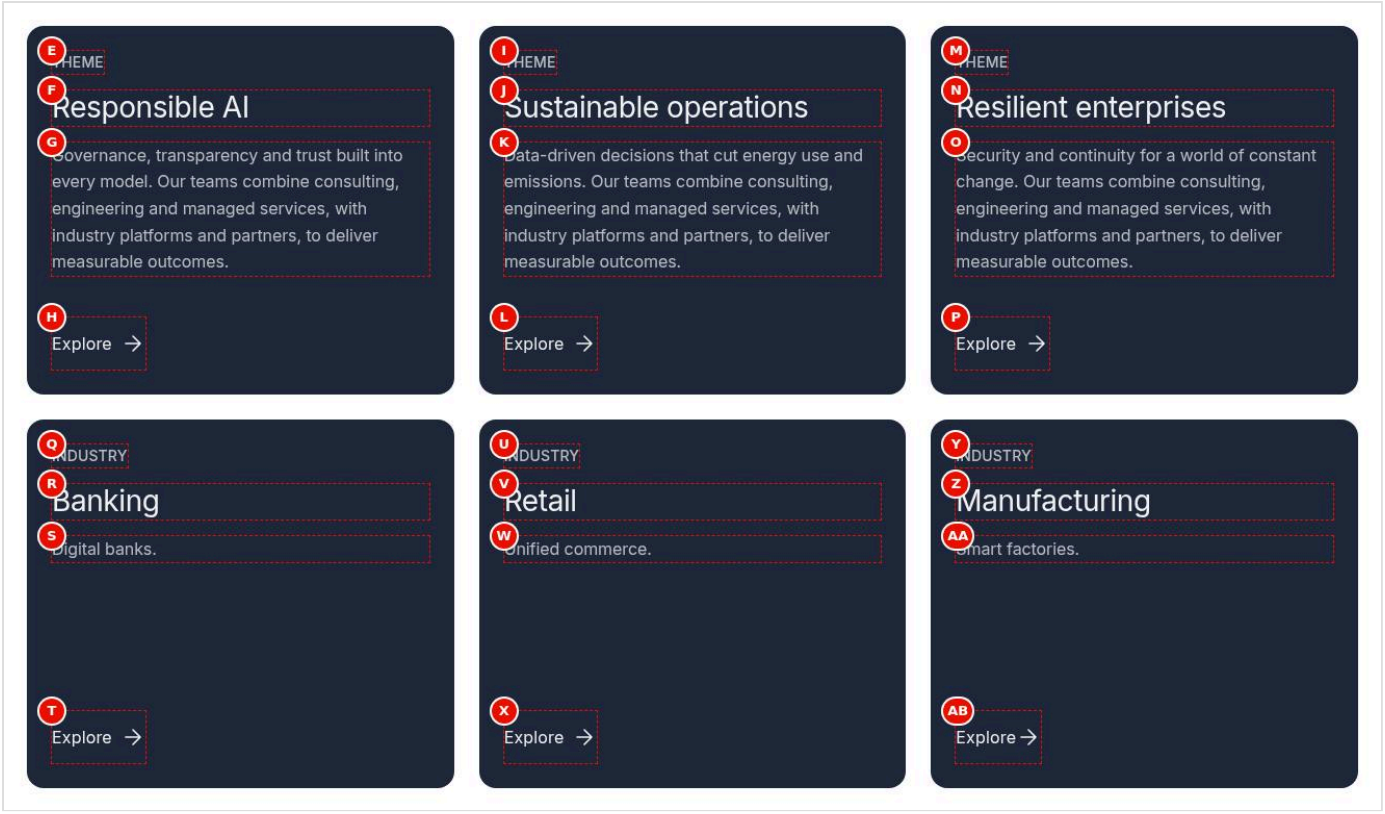


Figure: Equal card height — At 1440 px six dark #20293c panels (Panel height Content) in two rows of three with the same height, although the second row ("Banking", "Retail", "Manufacturing", with one-line descriptions) has much shorter texts than the first. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-equal-height.jpg) (handover screenshot; rendering at the time it was taken).

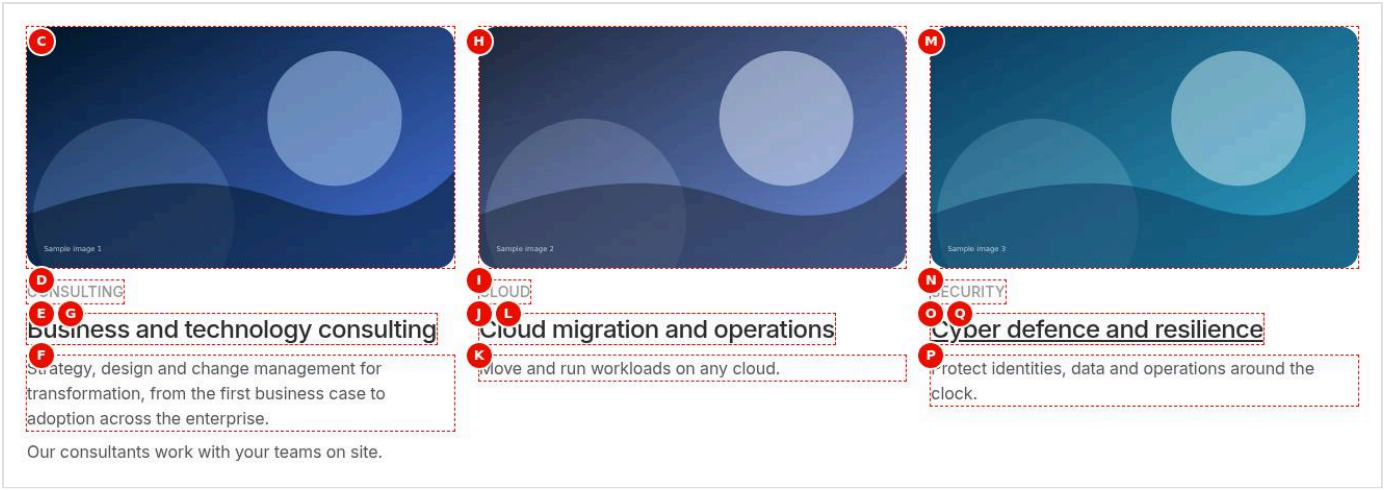


Figure: Cards not stretched — At 1440 px three image cards in a 3-column row ("Business and technology consulting", "Cloud migration and operations", "Cyber defence and resilience") of different heights, aligned to the top of the row. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-no-stretch.jpg) (handover screenshot; rendering at the time it was taken).



Figure: Unlinked cards — At 1440 px four compact (9:10) light grey statistic cards ("4bn", "600,000+", "150", "98%") without any link or button; the pointer rests on the second card and shows no hand cursor or link styling. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-unlinked.jpg) (handover screenshot; rendering at the time it was taken).

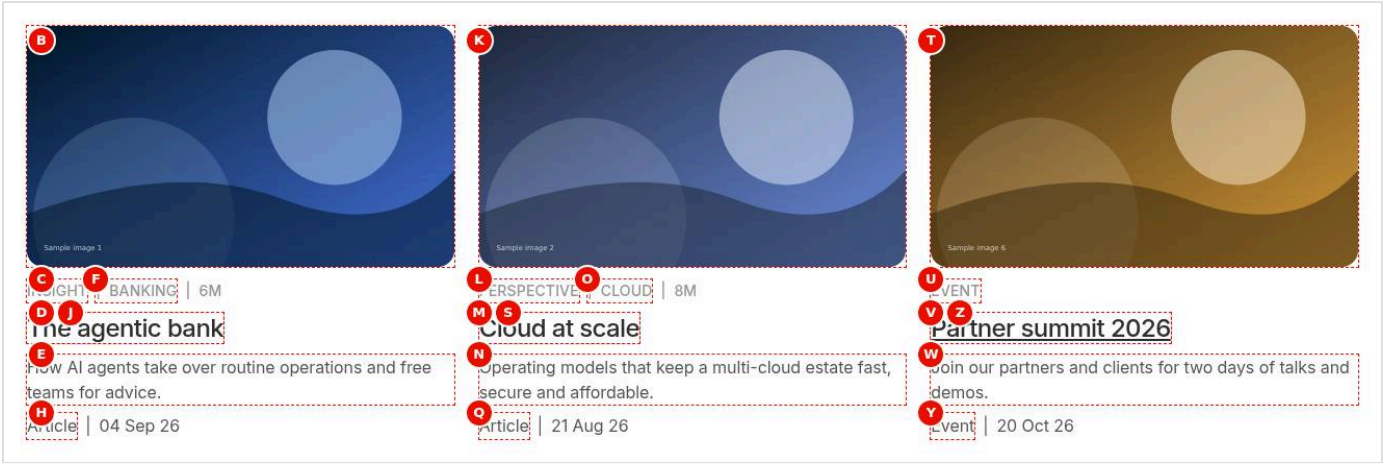


Figure: New-tab links — At 1440 px three image cards in a 3-column row; the pointer is on the third card's title "Partner summit 2026" (Event, "20 Oct 26"), which links to https://www.example.com/partner-summit in a new tab; the browser status bar isn't part of the page capture. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-new-tab.jpg) (handover screenshot; rendering at the time it was taken).

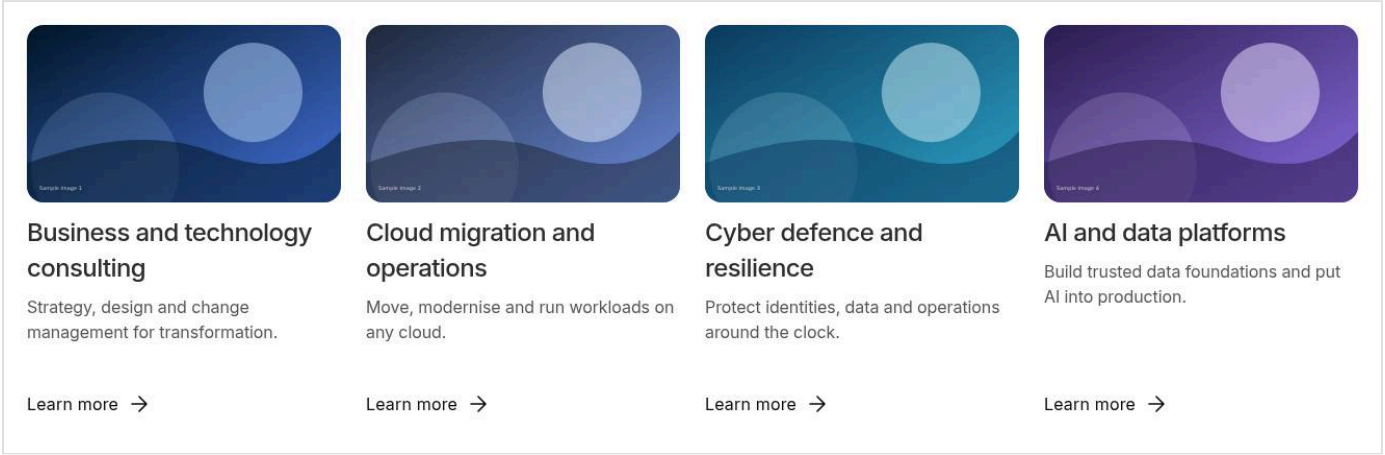


Figure: Legacy cards in the new design — At 1440 px an older Cards block (image and rich text cells, no block settings) rendered in the new design as four image cards in a row: each heading ("Business and technology consulting", ...) as the title, the sentence as description and "Learn more" as the call to action text link. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-legacy.jpg) (handover screenshot; rendering at the time it was taken).

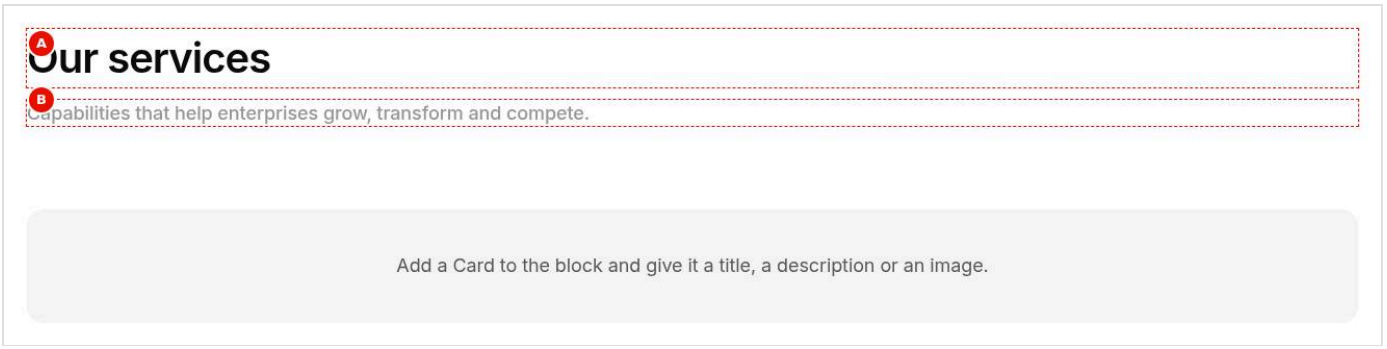


Figure: Empty state — In Universal Editor edit mode at 1440 px: the header "Our services" with its description, and below it a grey box with "Add a Card to the block and give it a title, a description or an image." (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/cards-empty-state.jpg) (handover screenshot; rendering at the time it was taken).

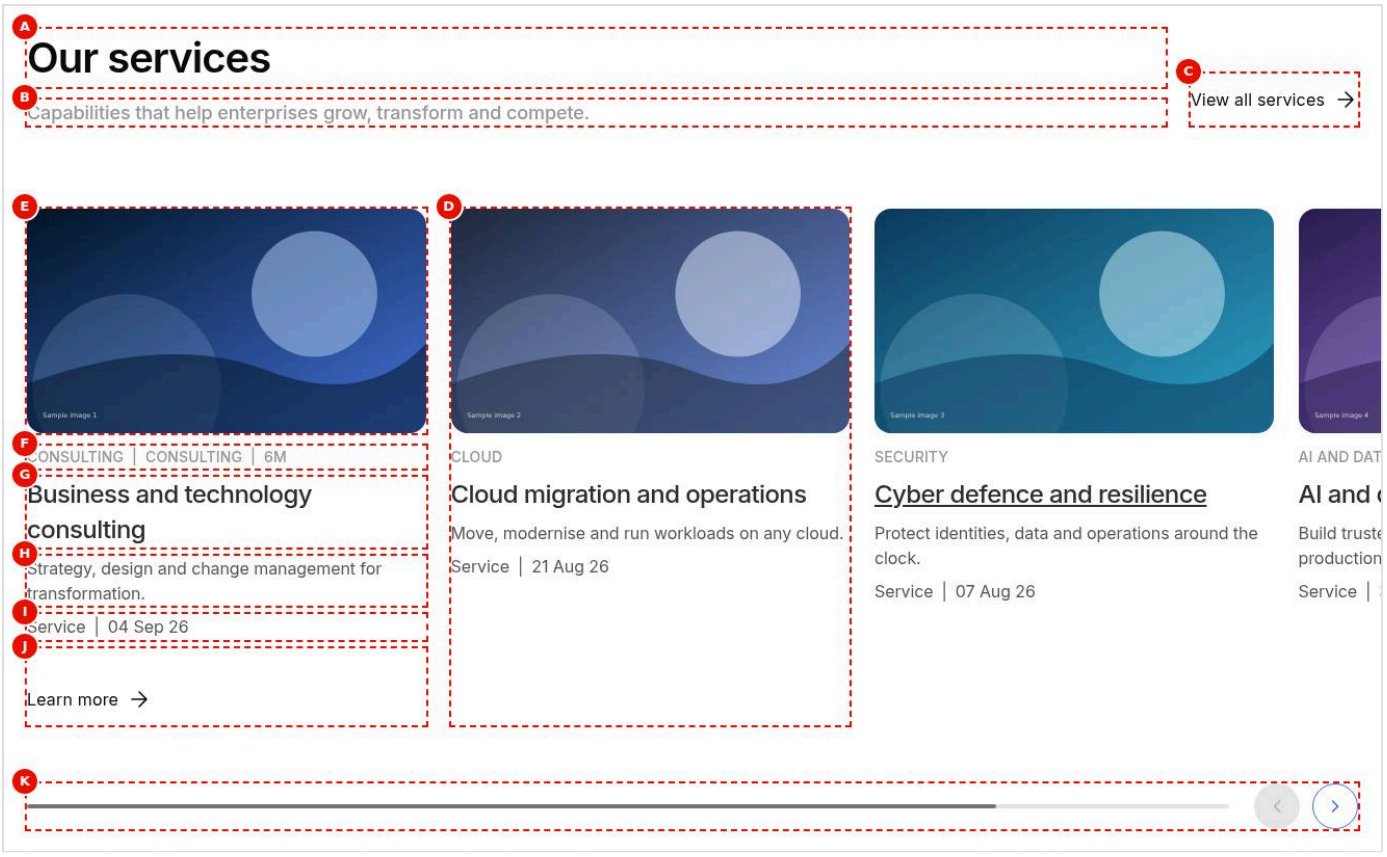


Figure: Universal Editor canvas — An annotated preview in editor mode at 1440 px (not a Universal Editor screenshot): the header "Our services" with description and "View all services" (A to C) and a rail of four service image cards with 3.2 visible (D to J), with buttons and indicator (K). The content tree with its four Card items (L) isn't part of the capture. (Annotated preview in editor mode with sample content; letters as in the annotations list, not a Universal Editor screenshot; file screenshots/cards-authoring-view.jpg) (handover screenshot; rendering at the time it was taken).

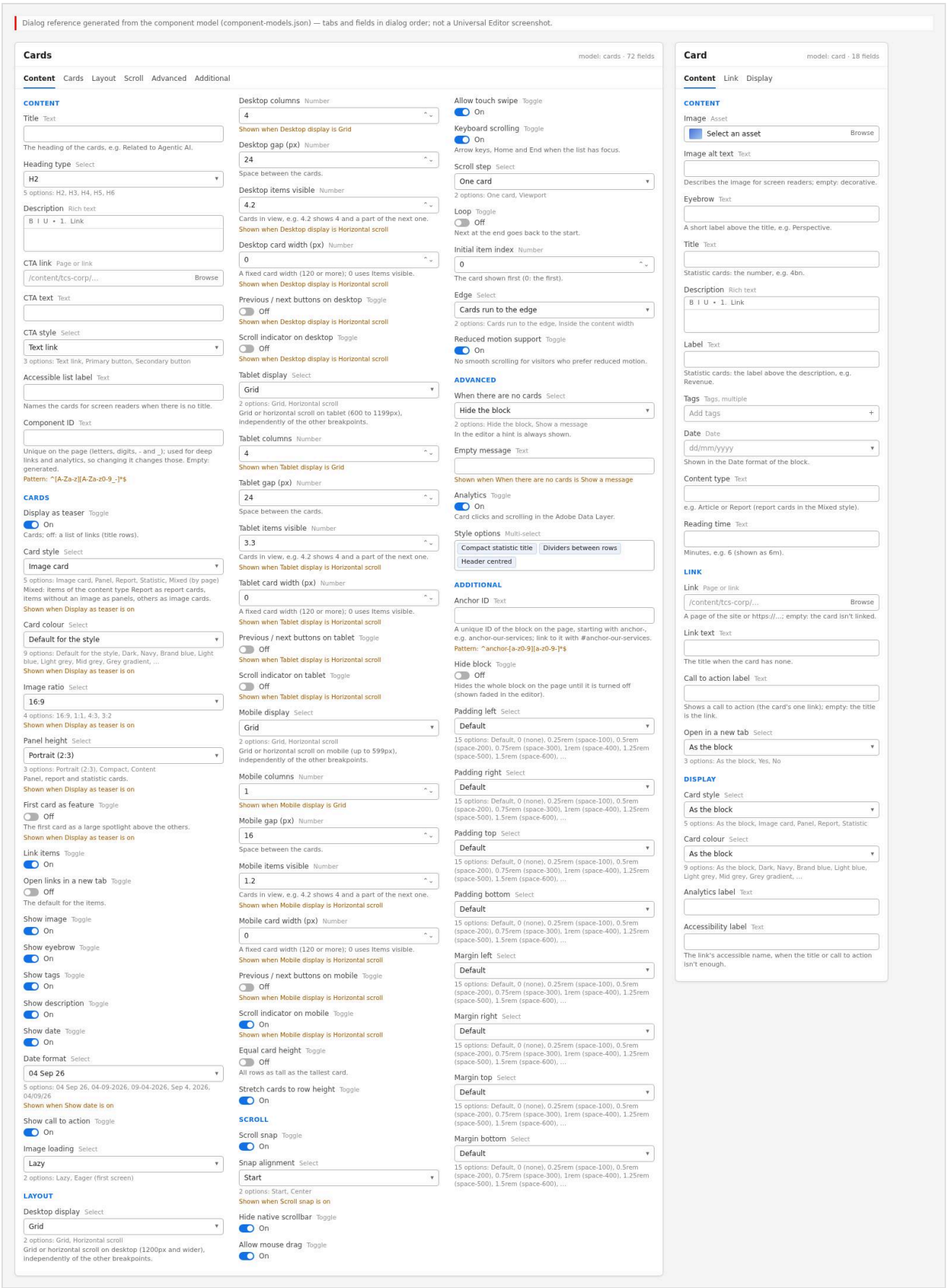


Figure: Properties panel — The Universal Editor properties panel of a selected Cards block showing the tabs Content, Cards, Layout, Scroll, Advanced and Additional, with the Cards tab open (Display as teaser, Card style, Card colour, Image ratio, Panel height, First card as feature, the Show switches, Date format, Image

loading). (Dialog reference generated from the component model, not a Universal Editor screenshot; file screenshots/cards-properties-panel.jpg) (handover screenshot; rendering at the time it was taken).

21. Known limitations

- Universal Editor dialogs were not captured: no author access in this program (see the access blockers).

22. Open items

- None.

23. Change history

Version	Date	Author	Change
1.0	8 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages